
Damping sets the limit: when linearised Vlasov theory predicts magnetised cross-beam energy transfer, and when it does not

Regime characterisation study
magnetised cross-beam energy transfer

Abstract

Cross-beam energy transfer (CBET) is routinely modelled in inertial-confinement-fusion (ICF) design with linear kinetic response functions, and that practice is now being extended to magnetised plasmas, where the ion-acoustic quasimode that mediates the transfer, the electromagnetic eigenmodes that carry it, and the polarisation selection rule that gates it are all modified by the field. We ask where this reduced description can be trusted. We reconstruct a linearised Vlasov magnetised-CBET model — full magnetised Maxwellian susceptibilities on unperturbed gyro-orbits, warm-fluid electromagnetic eigenmodes, a rank-one ponderomotive coupling integrated with the X/O dephasing over the interaction length, and the finite-box mode partition — and confront it with 160 kinetic reference points assembled from published particle-in-cell (PIC) gain measurements and from 34 new reduced-scale PIC simulations run here, spanning field strength, field orientation, electron temperature, laser intensity and polarisation. The verdict is not organised by magnetisation. It is organised almost entirely by a single quantity: the quality factor $Q = \Delta\lambda_{\text{res}}/\Gamma$ of the mediating low-frequency resonance. Where the mediating wave is strongly damped ($Q \lesssim 4.7$) the model reproduces measured gains with no adjustable quantity at all, to a median normalised discrepancy of 0.07–0.16, including the field-induced activation of transfer between orthogonally polarised beams — although the detuning at which that resonance sits is over-predicted by a factor 1.42–1.82 throughout, a separate and unexplained discrepancy. Where it is weakly damped ($Q \simeq 53$, the case of a carbon plasma with $ZT_e/T_i = 24$) the linear resonant amplitude gain reaches ~ 20 , far outside the undepleted-pump regime the model assumes, and the absolute prediction fails by more than an order of magnitude — and new simulations show the transfer there is nevertheless exactly linear in intensity ($I^{0.93}$ over a factor of eight), so saturation is not the explanation: the linear response function itself is wrong by a factor 37. Across all 86 cells with a measurable gain the over-prediction collapses onto $S = 0.30 Q^{1.12}$ ($\rho = 0.76$): the mediating resonance never behaves as though it were sharper than $Q_{\text{eff}} = 2.9$, whatever the linear damping rate says. A single split at $Q^* = 4.7$ classifies 91% of the cells. Magnetisation enters this picture only through its effect on Q : it lowers Q , and therefore restores predictivity, but only when the mediating wavevector is close to perpendicular to the field. The failure is attributed, under a pre-registered set of candidate mechanisms, to the assumption of a single strongly damped ion-acoustic mediator (Spearman $\rho = 0.68$, 95% CI [0.55, 0.78]); the competing comparison-validity explanations do not separate the classes. For ICF practice the operational statement is a pre-shot test on Q , computable from a single susceptibility evaluation, that costs nothing and decides whether a reduced CBET model or a kinetic simulation is required.

1 Introduction

Cross-beam energy transfer is the resonant exchange of energy between overlapping laser beams through a low-frequency plasma wave, and it is one of the few laser–plasma instabilities that ICF designers both exploit and fear. In indirect drive it is used deliberately, by detuning the wavelengths of inner and outer cones, to redistribute energy and control implosion symmetry [1, 2, 3]. In direct drive it is a loss channel that drains incoming light into outgoing scattered light and degrades hydrodynamic efficiency [4, 5, 6]. In both cases the quantity that enters the design loop is a *gain coefficient* evaluated from a linear kinetic response function; the whole edifice of ray-based CBET modelling rests on the assumption that a linearised treatment of the mediating ion-acoustic wave is quantitatively adequate [7, 8, 9].

That assumption is under pressure from two directions. First, it is known to fail in specific ways even in unmagnetised plasmas: ion trapping detunes the resonance and saturates the transfer [10, 11], ion heating and pump depletion cap the achievable gain [12], non-Maxwellian electron distributions shift the damping [13], kinetic and nonlinear effects modify the transfer at direct-drive conditions [14], and finite beam bandwidth and optical smoothing broaden the resonance far beyond its plane-wave width [15, 16]. Second, ICF experiments are increasingly magnetised, by imposed coils or by self-generated fields, with megagauss to tens-of-megagauss fields in the corona and near the laser entrance holes [18, 19, 20, 21, 22]. At those field strengths the electron cyclotron frequency is no longer negligible compared with the ion-acoustic frequency, the two electromagnetic eigenmodes are no longer degenerate, and a polarisation selection rule that forbids transfer between orthogonally polarised beams in an unmagnetised plasma is lifted [23, 25, 26, 27]. A magnetised linear CBET model now exists [26], and PIC simulations confirm its qualitative predictions [27], but the same simulations report quantitative differences that are not resolved.

The practical question is therefore not whether the reduced model is exactly right — it is not — but *where* it is right well enough to design with, and what changes when it is not. This paper answers that question by mapping the region of parameter space in which a reconstructed linearised Vlasov magnetised-CBET model reproduces kinetic reference data, and by identifying which of its assumptions gives way at each edge of that region.

The answer is unexpectedly compact. The validity of the reduced model is not controlled by the magnetic field strength, nor by the beam polarisation, nor by the laser intensity in the range explored. It is controlled by how strongly the mediating low-frequency wave is damped. Written as a quality factor $Q = \Delta\lambda_{\text{res}}/\Gamma$, the ratio of the resonant seed detuning to the full width of the resonance, the model is quantitatively predictive for $Q \lesssim 4.7$ and fails badly above it. The magnetic field matters only because it can move Q : near-perpendicular propagation of the mediating wave with respect to $\mathbf{B}$ introduces cyclotron phase mixing that broadens the resonance and *restores* predictivity, whereas parallel propagation leaves Q untouched.

2 The reduced model and what it assumes

2.1 Coupled-mode formulation

We reconstruct the linear model from standard magnetised kinetic theory, in the form in which it is used for CBET. Two transverse waves, a pump $(\omega_0, \mathbf{k}_0)$ and a seed $(\omega_1, \mathbf{k}_1)$, beat at $(\Delta\omega, \Delta\mathbf{k}) = (\omega_0 - \omega_1, \mathbf{k}_0 - \mathbf{k}_1)$ and drive a low-frequency density perturbation through the ponderomotive potential energy $\tilde{\Psi} = e^2 \tilde{A}_0 \tilde{A}_1^* (\hat{\mathbf{e}}_0 \cdot \hat{\mathbf{e}}_1^*) / (2m_e \omega_0 \omega_1)$. Retaining only the linear response of the electrons to $\tilde{\Psi}$ and of both species to the self-consistent field,

$$\frac{\delta \tilde{n}_e}{n_0} = -(k_a \lambda_{De})^2 X(\Delta\omega, \Delta\mathbf{k}) \frac{\tilde{\Psi}}{T_e}, \quad X \equiv \frac{\chi_e (1 + \chi_i)}{1 + \chi_e + \chi_i}, \quad (1)$$

with $k_a = |\Delta\mathbf{k}|$. The species susceptibilities are the linear magnetised Maxwellian ones, obtained by integrating the linearised Vlasov equation along unperturbed helical orbits,

$$\chi_s = \frac{1}{k^2 \lambda_{Ds}^2} \left[1 + \zeta_{s0} \sum_{n=-\infty}^{\infty} \Gamma_n(b_s) Z(\zeta_{sn}) \right], \quad \zeta_{sn} = \frac{\omega - n\Omega_s}{\sqrt{2} |k_{\parallel}| v_{ths}}, \quad b_s = \frac{k_{\perp}^2 v_{ths}^2}{\Omega_s^2}, \quad (2)$$

where $\Gamma_n(b) = I_n(b)e^{-b}$ and Z is the plasma dispersion function [33, 34]. The sum is truncated adaptively; for $b_s > 400$, where the cyclotron harmonics merge into a continuum, the unmagnetised Maxwellian form is used instead, a substitution verified to be smooth at the switch.

Substituting Eq. (1) into the transverse wave equation gives, for the complex *vector* seed envelope $\mathbf{A}$, a rank-one growth operator

$$\partial_x \mathbf{A} = i C X^* |\tilde{A}_0|^2 (\hat{\mathbf{e}}_0^* \cdot \mathbf{A}) \hat{\mathbf{e}}_0, \quad C = \frac{e^2 k_a^2}{8 m_e^2 \omega_0 \omega_1 c^2 k_1}. \quad (3)$$

The rank-one structure *is* the polarisation selection rule: only the component of the seed along the pump polarisation is amplified, so an unmagnetised plasma transfers no energy between orthogonally polarised beams. Magnetisation lifts the rule not by changing Eq. (3) but by changing the basis in which it acts.

2.2 Electromagnetic eigenmodes and the interaction length

The seed propagates as a superposition of the two electromagnetic branches of the magnetised plasma, which are non-degenerate. We obtain them from the warm-fluid magnetised dielectric tensor, $\epsilon = \mathbf{I} + i \sum_s (\omega_{ps}^2 / \omega) \mathbf{M}_s^{-1}$ with $\mathbf{M}_s = -i\omega \mathbf{I} - \Omega_s \mathbf{A} + i(\gamma T_s / m_s \omega) \mathbf{k} \mathbf{k}^T$, solving $\det[\epsilon - n^2(\mathbf{I} - \hat{\mathbf{k}} \hat{\mathbf{k}})] = 0$ for the two electromagnetic roots and taking the null vectors as the complex unit polarisations $\hat{\mathbf{e}}_{1,2}$ [24]. Thermal corrections to n^2 are of order $(k v_{te} / \omega)^2 \sim 10^{-3}$ here. Projecting Eq. (3) on this basis gives a two-branch system whose off-diagonal terms carry the X/O dephasing $\exp[i(k_{b'} - k_b)x]$; we integrate it over the interaction length rather than exponentiating a growth rate, because in the reference configuration $(k_1 - k_2)L_x$ is of order unity and the branches exchange energy as they dephase.

The measured observable is an energy gain, $g = \ln[U(t)/U(t_0)]$ with $U = \frac{\epsilon_0}{2} \int \hat{E}^2 dx$, so we report $g = \ln[L^{-1} \int_0^L |A|^2 dx / |A(0)|^2]$, which reduces to $\ln F(G)$, $F(G) = (e^G - 1)/G$, for a pure exponential envelope. Finally, a finite plasma of length L_x supports discrete wavevectors spaced by $2\pi/L_x$, and a seed entering at a given frequency generally excites the two adjacent box modes with weights $f_{\pm}$; we carry this partition through as an $f_{\pm}^2$ -weighted average of the two single-mode gains. The cross term vanishes because $\langle \cos(\delta k x) \rangle = 0$ for $\delta k = 2\pi/L_x$.

2.3 The three physical assumptions

Three approximations define the model and are the candidates for failure. **(i)** The drive is the warm-fluid, electrostatic-like ponderomotive potential evaluated at a single averaged frequency, with the magnetisation correction and the frequency-derivative terms dropped; this is justified only inside the two-sided window $\Delta\omega \ll \omega_{ce} \ll \omega_0$. **(ii)** The ion response is linear and ponderomotive-free: only the linear magnetised ion susceptibility is kept, so the ion quiver motion is neglected. **(iii)** The transfer is mediated by a *single, strongly damped* ion-acoustic resonance, with the perturbed distribution integrated along unperturbed gyro-orbits.

Assumption (iii) is the one that will matter, and it is worth making explicit what it buys. A strongly damped mediator has a broad resonance, so the gain is a smooth function of detuning, it is insensitive to exactly where the resonance sits, and — crucially — its peak amplitude is bounded, which keeps the transfer inside the undepleted-pump regime in which the linearisation was performed. A weakly damped mediator has none of these properties.

2.4 What the reconstruction reproduces

Before any quantitative comparison we checked that the reconstruction is informative, i.e. that it reproduces the qualitative magnetisation phenomenology that motivates the magnetised model at all. We turned the published qualitative statements into 10 machine-checkable tests (logged in `logs/qualitative_reproduction.json`) and the reconstruction passes 8 of them: with increasing $|B|$ the parallel-polarisation gain falls at all three orientations; the orthogonal-polarisation gain rises from exactly zero when $\Delta \mathbf{k} \parallel \mathbf{B}$ and stays near zero when $\Delta \mathbf{k} \perp \mathbf{B}$; the gain curve shifts and broadens at high $|B|$ in the perpendicular case; the X and O branches separate when $\Delta \mathbf{k} \perp \mathbf{B}$; and the two polarisation curves cross, at smaller $|B|$ for oblique than for parallel $\Delta \mathbf{k}$. Two tests

fail, and both are threshold artefacts rather than qualitative disagreements: for $\Delta\mathbf{k} \parallel \mathbf{B}$ the branch split at 10 MG is 0.016, which is below the 0.02 we required but also below the measurement's own resolution there (the split only becomes clear above 30 MG, where the model tracks it well — see Sec. 5.4); and the two polarisation curves approach each other at 64 MG (0.86 against 0.65) without strictly crossing, which the measurement also does not do at that field (Fig. 2).

3 Kinetic reference data

3.1 Published measurements

The reference set is the processed output of 2D-3V PIC simulations of the same interaction, made available with the study that reported them [27, 28]. It contains gain curves $g(\Delta\lambda)$ and eigenmode-resolved gains for a fully ionised carbon plasma at $n_e = 3.5 \times 10^{20} \text{ cm}^{-3}$, $T_e = 4T_i = 0.8 \text{ keV}$, crossing angle 30° , with $\mathbf{B}$ in the crossing plane at three azimuths — corresponding to $\theta_{kB} \equiv \angle(\Delta\mathbf{k}, \mathbf{B})$ of 0° , 45° and 88° — and $|B|$ from 0 to 64 MG; and gain curves for a hydrogen plasma at $n_e = 3 \times 10^{20} \text{ cm}^{-3}$, $T_e = 4T_i \in \{0.4, 1.2\} \text{ keV}$, crossing angle 24° , with $\mathbf{B}$ out of the crossing plane ($\theta_{kB} = 71^\circ$). We use these files exactly as distributed, and we use *all* of them: the archive's $\theta_{kB} \simeq 45^\circ$ scan is denser than the field values declared in advance, carrying 6 additional strengths (12, 14, 16.5, 20, 24, 28 MG), and those cells are included rather than trimmed back to the declared list. Conversely, the number of declared field values missing at any orientation is 0: the grid actually analysed is a strict superset of the one registered in advance, and nothing was dropped from it. Following the protocol under which they were produced, the common background offset g_0 — the negative gain measured at $B = 0$ with orthogonal polarisations, where linear theory predicts none — is subtracted from every gain of the same family; for carbon $g_0 = -0.39$, which reproduces the published figures.

3.2 New reduced-scale simulations

The published set does not vary laser intensity, does not place the three temperatures in one geometry, and covers only three field orientations. We therefore ran 41 new EPOCH [32] simulations using the authors' own input deck, changing only the user-facing constants. The reduced configuration — fixed in advance and held identical at every sweep point — is $L_x = 200\lambda_0$ (against $500\lambda_0$), $L_y = 10\lambda_0$ unchanged, 15 cells per λ_0 (against 20) and 16 particles per cell (against 20), with the same end time in units of the seed transit time and the same three output times, so that the t_0, t_1, t_2 gain protocol maps over unchanged. Runs use 32 MPI ranks each on disjoint physical cores.

Three new arms extend the map: an intensity arm at $I \in \{0.25, 0.5, 1.0, 2.0\} \times 10^{15} \text{ W cm}^{-2}$ (pump and seed kept equal), a temperature arm at $T_e \in \{0.4, 0.8, 1.2\} \text{ keV}$ in the carbon reference geometry, and an orientation arm at two azimuths the published set does not cover ($\theta_{kB} = 25^\circ$ and 70°). Numerical checks at 12, 15 and 20 cells per λ_0 and at $L_x = 200$ and $300\lambda_0$ bracket the reduction, and a seed ensemble at two boundary-critical points measures the run-to-run scatter of the PIC noise realisation. EPOCH seeds its generator deterministically; a one-line patch exposes the base seed through the environment so that independent realisations are reproducible.

3.3 Analysis

New runs are processed with a NumPy port of the published MATLAB analysis chain: the same Gaussian bandpass filter around $k_x = \pm k_0$, $k_y = 0$ with the same widths, the same transverse integration at $f_y \in \{1, 0.75, 0.5\}$, the same Faraday-rotation phase shift $\hat{k}\Delta x = \psi$ before the eigenmode combination, and the same gain definition at the same three dumps. One deviation was necessary and is used uniformly: the reference takes the common wavevector as the mean of the peaks of the two transverse components, which is meaningless when one component carries no seed — exactly the case for a linearly polarised seed in a weakly magnetised plasma, where the subsequent narrow 1D filter then removes the seed entirely. We take the common wavevector from the peak of the combined transverse power spectrum instead.

Deviations from the source framework, in full. Four, all disclosed and all applied uniformly. (i) The analysis chain is a NumPy port of the published MATLAB package rather than the package itself, because no MATLAB or Octave interpreter was available; the port is checked against the

properties the package’s own unit tests assert. (ii) Within that port, the common wavevector used for the eigenmode combination is taken from the combined transverse power spectrum rather than from the mean of the two component peaks, for the reason given above. (iii) The electromagnetic eigenmode polarisation vectors, which the published driver imports from a separate repository that is not distributed with it, are reconstructed here from the same warm-fluid dielectric tensor. (iv) The published input deck specifies `simple_laser/open` boundaries with the CPML thickness commented out, while the accompanying text describes CPML; we follow the deck, because the deck is the artefact that produced the reference data. Everything else — plasma model, reference parameters, launch geometry, gain definition, dump times, uncertainty bands — is unchanged.

4 Comparison protocol

The protocol was fixed in advance of the comparison and is recorded, with a timestamp, in the study’s plan file; the two amendments made afterwards are recorded and dated in the same place and are described below.

Observable. The near-peak CBET energy gain. On the measurement side this is g at the detuning at which the reference study operates, $\Delta\lambda = 0.9 \text{ \AA}$ for carbon (the measured curves peak at $0.89\text{--}0.96 \text{ \AA}$) and the nearest available grid detuning, $1.0 \text{ \AA}$, for hydrogen. On the model side it is g at the detuning that maximises the model’s own low-frequency response, because the model’s resonance does not sit at the same place — an offset we quantify rather than register away.

Agreement measure and verdict. $D = |g_{\text{model}} - g_{\text{PIC}}|/g_{\text{ref}}$, where g_{ref} is the offset-corrected unmagnetised parallel-polarisation gain measured in the same family. Normalising by a fixed positive number per family, rather than by g itself, keeps D meaningful for the orthogonal-polarisation cells where g passes through zero. A cell *holds* if $D \leq 0.30$; it *breaks* if $D > 0.30$ and the discrepancy also exceeds the measurement uncertainty σ ; otherwise it is *inconclusive*. The threshold 0.30 was declared before any residual was computed, together with a pre-declared robustness band at 0.20 and 0.40 ; changing it over that band changes the number of holding cells only from 53 to 63 out of 160 , so no conclusion here depends on it.

Uncertainty. σ combines the published $t_1\text{--}t_2$ and f_y spread of the cell, a 0.10 systematic on the background offset, and — for new runs — the measured seed-to-seed scatter.

Candidate failure modes. Seven were declared in advance, four of them *comparison-validity prerequisites* whose failure means the comparison did not test the physics (C1 loss of quasi-steady state; C2 pump depletion; C3 sheath fields and stray light; C4 unresolved finite-domain mode structure) and three of them *physical assumptions* of the model (V1 the two-sided drive window; V2 the neglected ion ponderomotive response; V3 the single strongly damped mediator). Each carries an indicator and an *a priori* threshold, and the attribution rule — comparison-validity wins over physical when both fire; “unattributed” when two of a kind fire or none does — was fixed with them.

One post-hoc addition, tested on fresh cells. The mechanism that turns out to matter, V3, was declared in advance. The *coordinate* in which the boundary is sharpest — the resonance quality factor Q — and the value at which it splits were not: they emerged from the residuals. Because the published data samples Q at only two clusters ($\simeq 3$ and $\simeq 50$), a split placed in the gap between them would be a classification rather than a measured transition. We therefore added an arm of new simulations at cells that took no part in the discovery: the carbon reference plasma with the ion temperature raised at fixed T_e , so that ZT_e/T_i becomes 9 and 6 instead of 24 and Q lands at 16.1 and 7.2 , inside the gap. The model’s absolute predictions for those four cells were computed and written to `logs/blind_prediction_TI.json` before the simulations were run. This adds cells to the grid and removes none.

Two amendments. (A1) The reduced PIC configuration was shortened once, from 250 to $200\lambda_0$ and from 20 to 16 particles per cell, after the first three runs revealed that concurrent MPI jobs on this node were binding to overlapping cores and the declared sweep would not have finished; those three runs were discarded unanalysed. (A2) While building the model, and before any magnetised residual was computed, we found that its absolute prediction at the carbon reference point is $g = 36.4$ against

a measured 1.51. We therefore report the model in two forms: *absolute*, exactly as declared, which is the primary form; and *registered*, in which the amplitude gain of every cell is multiplied by one scalar κ per interaction-length scale, fixed by the unmagnetised parallel-polarisation gain of the carbon reference point ($\kappa = 0.063$ for the published $500\lambda_0$ runs). Every dependence on field, orientation, temperature, intensity, polarisation and branch remains unadjusted in both forms.

5 Results

5.1 The boundary is one-dimensional, and it is not the magnetic field

Figure 1 shows the verdict at all 160 cells. Panel (a) plots it in the plane the study was designed around — field strength against the orientation of the mediating wavevector — and the pattern there is not clean: cells that hold and cells that break interleave at almost every field strength, and at $\theta_{kB} \simeq 0^\circ$ and 45° essentially everything breaks regardless of $|B|$.

Panel (b) plots the same verdicts against the quality factor of the mediating resonance, $Q = \Delta\lambda_{\text{res}}/\Gamma$, computed by the model from Eq. (2) alone. The map collapses. Every hydrogen cell, at both temperatures and at every field strength, has $Q \simeq 3.0$ – 3.3 and holds, with median D of 0.07 and 0.16. Every carbon cell whose resonance remains narrow has $Q \simeq 53$ and breaks, with median $D = 16.5$. A single split at $Q^* = 4.7$ classifies 91% of the cells correctly. Below the split the median D is 0.08; above it, 15.0 — a contrast of more than two orders of magnitude.

The physical content is direct. In the carbon reference plasma $ZT_e/T_i = 24$, ion Landau damping of the mediating wave is negligible, and the resonance the model predicts is $0.024 \text{ \AA}$ wide at $\Delta\lambda_{\text{res}} = 1.26 \text{ \AA}$. Its peak amplitude gain is ~ 20 : a plasma that would be violently unstable and completely pump-depleted at resonance. The measurement, sampled on a $0.3 \text{ \AA}$ grid, never resolves that spike and instead returns a smooth hump of amplitude gain $\simeq 2.5$ and width $\simeq 1.3 \text{ \AA}$. The model and the measurement are not describing the same object. In hydrogen, where $ZT_e/T_i = 4$, ion Landau damping is strong, the predicted resonance is $0.36 \text{ \AA}$ wide at $1.32 \text{ \AA}$, the peak linear gain is of order unity, and model and measurement agree point by point across the whole field scan (Fig. 2, lower panels) with no adjustable quantity whatsoever — including the rise of the orthogonal-polarisation gain from zero to $\simeq 0.8$, which is the distinctive magnetised prediction.

5.2 Magnetisation enters only through the damping

Panel (c) of Fig. 1 explains why the magnetic field appears in the map at all. When $\Delta\mathbf{k}$ is nearly parallel to $\mathbf{B}$ the mediating wave is effectively unmagnetised and Q is flat in $|B|$: the field changes the lasers, not the mediator, and the model’s absolute failure persists to 64 MG. When $\Delta\mathbf{k}$ is nearly perpendicular, $k_{\parallel}$ shrinks, the electron parallel phase velocity moves into the region where Landau damping is strong, the finite-Larmor-radius factor $\Gamma_0(b_e)$ cuts the electron response, and Q falls monotonically from 53 at $B = 0$ to 2.5 at 50 MG, crossing Q^* between 5 and 8 MG. Exactly there, the carbon cells begin to hold: at $\theta_{kB} \simeq 88^\circ$ and $|B| \geq 10 \text{ MG}$ the absolute model reproduces the measured gain to $D = 0.06$ – 0.21 , for both polarisations (Table 1). Magnetisation does not break the reduced model; where it acts on the mediator, it repairs it.

The one exception is instructive. At $\theta_{kB} \simeq 88^\circ$ and 32 MG the model develops a narrow additional resonance at low detuning, Q jumps from 2.6 to 353, and that cell breaks while its neighbours at 16 and 50 MG hold. This is the emergence of a weakly damped magnetised mode the single-mediator picture does not represent, and it is the fingerprint the pre-registered V3 indicator was written to catch.

5.3 After registration: what the magnetisation physics gets right

The absolute failure in carbon is an overall-amplitude failure, and it is worth asking separately whether the *shape* of the magnetisation response survives it. Panel (d) of Fig. 1 and the carbon panels of Fig. 2 show the registered form, in which one scalar per interaction-length scale is fixed at $B = 0$. It does survive, and well: 112 of 160 cells hold. The model reproduces the onset field, the depth of the parallel-polarisation reduction, the activation and saturation of the orthogonal-polarisation gain, and the crossing point of the two polarisations at $\theta_{kB} = 0^\circ$ and 45° across three decades of field

strength. The registered form fails in two places: at $\theta_{kB} \simeq 88^\circ$ between 1 and 10 MG, where the model predicts a collapse of the gain far steeper than measured, and at the very top of the field range at $\theta_{kB} = 45^\circ$, where the measured gain turns back up and the model does not.

5.4 The eigenmode-resolved secondary observable

The published data resolves the gain into the two electromagnetic branches, and the model predicts them separately, so the branch split $g_X - g_O$ is an independent test of the polarisation physics that the total gain averages over. The registered model reproduces it well where the field acts on the light waves along the mediating wavevector: at $\theta_{kB} \simeq 0^\circ$ the measured split is $+0.13$ at 32 MG and $+0.67$ at 64 MG against a predicted $+0.11$ and $+0.54$. It fails at intermediate orientation: at $\theta_{kB} \simeq 45^\circ$ the measurement develops a split of $+0.56$ at 64 MG while the model gives $+0.03$. At $\theta_{kB} \simeq 88^\circ$ both are consistent with zero (-0.02 measured, -0.00 predicted), which is itself a correct prediction. The pattern — branch physics right when $\Delta \mathbf{k} \parallel \mathbf{B}$, absent when $\Delta \mathbf{k} \perp \mathbf{B}$, and under-produced in between — points at assumption (i): the drive is evaluated with an unmagnetised electron quiver response, so the only route by which the two branches can gain differently is their polarisation overlap with the pump and their mutual dephasing, and at oblique angles that route alone is not enough.

5.5 Mechanism attribution

Figure 3(a) gives the association of each pre-registered indicator with the disagreement metric, and panel (b) asks the sharper question: does crossing the indicator’s *a priori* threshold actually separate holding cells from breaking ones? The seven *p*-values are Benjamini–Hochberg corrected at FDR 0.05 (critical $p = 0.0429$).

Only V3 does both. It has the largest rank correlation, $\rho = 0.68$ [0.55, 0.78], and crossing its threshold moves the median D from 0.08 to 15.0, with the fraction of breaking cells going from 5% to 72%. C2, pump depletion, correlates significantly ($\rho = 0.24$) but does *not* discriminate: its indicator is above threshold in the hydrogen families too, where the model works, so 68% of cells above its threshold break against 58% below — a real but non-separating association, and a reminder that a mechanism can be present without being the binding constraint. C3 and C4 never cross their thresholds anywhere in the swept region (their indicators peak at 0.259 and 0.054 against declared thresholds of 0.35 and 0.25), so although C4 in particular correlates strongly with D it is never *admissible*: its correlation is carried by the same carbon/hydrogen contrast that Q explains, not by anything crossing. C1 crosses in 19 cells, all of them new reduced-scale runs where the smaller signal makes the t_1 – t_2 and f_y spread a larger fraction of the gain; it is a statement about the new runs’ measurement precision rather than about the theory. V2 crosses in exactly 3 cells — the three at $2 \times 10^{15} \text{ W cm}^{-2}$, where the *electron* beat ponderomotive potential reaches $0.058 T_e$ — while the ion term it actually names stays at $4 \times 10^{-4} T_i$ throughout, suppressed by $Z^2 m_e/m_i$. Two of those three cells break and one holds, so even where it crosses it does not separate. V1, the drive window, is the one indicator with no association at all ($\rho = -0.06$ [−0.23, 0.12]), which is a clean negative result: over 0–64 MG the two-sided window $\Delta\omega \ll \omega_{ce} \ll \omega_0$ is never the binding constraint. Its indicator diverges at $B = 0$, where ω_{ce} vanishes; we rank those cells as the largest value rather than dropping them, which is the faithful ordinal treatment, while noting that the correction assumption (i) discards is identically zero at zero field, so the divergence is definitional rather than physical.

It matters to separate what the pre-registered *rule* labels each cell as from what the evidence supports, because here they differ. The rule gives comparison-validity precedence whenever such a mechanism is admissible, and C2’s indicator is admissible wherever the gain is appreciable, so by construction the rule labels 46 of the 100 breaking cells C2, 33 V3, 12 C1 and 9 unattributed — 62 of them carrying the “comparison-compromised” flag. We report those counts because they are what the declared procedure produces, but they are not the conclusion. C2 and V2 are refuted by direct measurement in the new runs (Sec. 5.6): the transfer is linear in intensity, the driven density perturbation is linear in intensity, and the measured pump-band loss has median magnitude 2%. A mechanism that is admissible everywhere and refuted where it is testable is not the binding constraint, and the label it collects under a precedence rule should not be read as evidence for it. The attribution supported by the evidence is V3, the assumption of a single strongly damped ion-acoustic mediator, and it was stated in advance. We note the honest caveat that the *threshold* attached to it in advance, $Q > 3$, is too tight to discriminate: the hydrogen families sit at $Q \simeq 3.0$ – 3.3 and hold. The discrimination is

carried by the effect size, not by the pre-declared cut, and we report it that way; the data-driven split at $Q^* = 4.7$ is quoted as a descriptive statistic, not as a hypothesis test.

5.6 What the intensity arm rules out, and what it leaves

The absolute over-prediction in the weakly damped regime has an obvious candidate explanation — the predicted resonant gain is so large that the real transfer must be saturating — and the new intensity arm rules it out. Over a factor of eight in intensity the measured amplitude gain scales as $I^{0.93}$ at zero field and $I^{1.00}$ at 10 MG, the driven density perturbation scales linearly with intensity as well, and the measured pump-band energy change stays at a median 2%. A saturated transfer would do none of these. The transfer is in the linear regime, and the linear response function is nevertheless wrong: it over-predicts the driven density perturbation by a factor 15 and the amplitude gain by a factor 37 at the same operating point.

That is a stronger statement than “the model breaks”, and it constrains what can be responsible. Any explanation must (i) be linear in the drive, (ii) suppress the response by a factor that grows with Q , and (iii) leave the strongly damped families essentially untouched. A universal instrument function — finite beam aperture, finite interaction time, anything that convolves the resonance with a fixed width W — satisfies (i) and (ii) but not (iii): the width needed to suppress the carbon response by 37 is $W \simeq 0.85 \text{ \AA}$, and the same width would suppress the hydrogen response by 3.4, whereas the observed hydrogen suppression is 1.1. Whatever limits the response of a weakly damped mediator therefore acts selectively on narrow resonances in a way a fixed convolution does not reproduce.

It does, however, follow a law — and we flag at once that this law was *not* pre-registered. The mechanism it sharpens, V3, was; the functional form below is a description of the residuals, arrived at after seeing them. It is presented as an empirical fit, not a derivation, and the four confirmatory cells that pin it down in the intermediate- Q region were run before it was written, on parameters chosen only to fill the Q gap. Writing $S = G_{\text{model}}/G_{\text{PIC}}$ for the over-prediction of the amplitude gain, the 86 cells with a measurable positive gain on both sides collapse onto

$$S = 0.30 Q^{1.12}, \quad \text{Spearman } \rho = 0.76, \quad (4)$$

across a decade and a half in Q and in S , and across published and new simulations, carbon and hydrogen, magnetised and not (Fig. 4c). The exponent is unity to within the fit, so Eq. (4) says something simple: the *effective* width of the mediating resonance is not the linear damping width but a fixed fraction of the resonant detuning. Dividing, Q/S is constant at $Q_{\text{eff}} = 2.9$ (Fig. 4d). The mediating low-frequency resonance never behaves as though it were sharper than $Q \simeq 3$, whatever the linear damping rate says, and the linear model is predictive exactly when its own Q is at or below that ceiling. That is the failure of assumption (iii) — one broad, strongly damped mediator — stated quantitatively, and it converts the verdict from a yes/no into a correction factor.

The ceiling is not universal: it depends on the interaction length. Fitted separately, the published $500\lambda_0$ configuration gives $S = 0.32 Q^{1.02}$ over 62 cells with $Q_{\text{eff}} = 3.0$, while the $200\lambda_0$ runs give $Q_{\text{eff}} = 1.9$. The convergence arm measures the same thing directly: holding everything else fixed and lengthening the domain from 200 to $300\lambda_0$ raises the gain from 0.209 to 0.455, i.e. an amplitude gain scaling as $L^{1.83}$ rather than the L^1 a constant spatial rate would give. A super-linear length scaling and a length-dependent ceiling are the same statement: the response of a weakly damped mediator is limited by a coherence length, and a longer interaction region recovers more of the linear resonance. The exponent at the published length is 1.02, i.e. unity within the fit, so at fixed length the effective width really is a fixed fraction of the resonant detuning. Values of Q_{eff} quoted below refer to the $500\lambda_0$ configuration, which is the closer of the two to ICF interaction lengths.

One cell is excluded from the fit and shown separately: at $\theta_{kB} \simeq 88^\circ$ and 32 MG the resonance search lands on a narrow secondary magnetised mode with $Q = 353$, where the quantity called Q is no longer the width of the resonance that mediates the transfer, and S is 2.8 rather than the ~ 100 Eq. (4) would give.

5.7 A residual the model cannot absorb: the resonance sits in the wrong place

The comparison above is between two near-peak gains: the measurement at the detuning its own curves peak at, the model at the detuning its own response peaks at. Those two detunings are not the same, and the difference is systematic. At $B = 0$ the measured gain curve peaks at $0.89 \text{ \AA}$ for carbon,

0.81 Å for hydrogen at 0.4 keV and 1.25 Å at 1.2 keV, while the model's resonance sits at 1.26, 1.32 and 2.28 Å — a factor 1.42 to 1.82 too high, i.e. the measured phase velocity of the mediating wave is only 0.55–0.71 of the linear kinetic prediction at the same wavenumber.

This is not the amplitude problem in disguise: it survives in the hydrogen families where the amplitude agrees, and it is present at every field strength. A plane-wave linearised model cannot produce it — the ion-acoustic root of Eq. (2) is what it is. The obvious candidates are a nonlinear frequency shift from ion trapping, which is known to detune CBET downwards in exactly this way [10, 11], the finite angular content of the two beams in a $10\lambda_0$ -tall box, which spreads the mediating wavenumber by about $\pm 20\%$, and ion heating during the interaction. We can exclude neither with the diagnostics run here, and we do not fold the offset into the comparison: it is reported as an outstanding discrepancy that any user of a linear CBET model at these conditions should expect.

The *shift* of the peak with field, which the source study reports qualitatively, is not a discriminating test at this sampling. Between $B = 0$ and 10 MG at $\theta_{kB} = 45^\circ$ the measured peak moves from 0.89 to 0.96 Å and the modelled resonance by 0.001 Å; both are far below the 0.3 Å spacing of the measured detuning grid, so the comparison is dominated by the parabolic interpolation used to locate the peak. We record it as untestable with the available data rather than as agreement.

5.8 Compactness

Figure 4(a) ranks the swept axes by the fraction of the rank variance of D each explains. The mediating-resonance damping leads ($\eta^2 = 0.49$), followed by the two physical routes to it: the ion Landau damping parameter ZT_e/T_i (0.44) and the orientation of the mediating wavevector (0.34). Electron temperature at fixed ZT_e/T_i follows at 0.32; the new carbon temperature arm is what separates it from Q , because the published data varies T_e and species together. The three axes the magnetised problem is usually organised around are the inert ones: polarisation 0.12, laser intensity 0.06, and field strength 0.04. The boundary is, to a good approximation, a one-dimensional surface in a derived coordinate rather than a four-dimensional surface in the control parameters — which is the useful outcome, because Q is computable in advance from a single susceptibility evaluation.

Figure 4(b) shows why the reduction works, by plotting the mediating response itself: a spike two hundredths of an Ångström wide for unmagnetised carbon, against half-Ångström-wide resonances for hydrogen and for near-perpendicular magnetised carbon.

5.9 The new arms

Figure 5 reports the reduced-scale simulations.

Validation. Panel (a) compares the new $200\lambda_0$ field scan at $\theta_{kB} = 45^\circ$ with the published $500\lambda_0$ scan. The unmagnetised amplitude gain falls from $G = 2.52$ to 0.44, a ratio of 0.17 where proportionality to interaction length would give 0.40: the reduction costs more gain than its length alone accounts for, by a factor 2.3. After that single rescaling the reduced-scale points reproduce the published field dependence at the 4 common field strengths to a mean 9% of the unmagnetised gain (largest deviation 20%). The reduction is therefore informative about the shape of the magnetisation response, which is what the sweep uses it for, but not about its absolute size; every new-run verdict is referred to the reduced-scale unmagnetised gain for this reason.

Intensity. Panel (b) tests the one scaling the linearised model asserts exactly, proportionality of the amplitude gain to the pump intensity: at $B = 0$ MG the amplitude gain scales as $I^{0.93}$ over a factor 8 in intensity (linear theory: I^1); and at $B = 10$ MG the amplitude gain scales as $I^{0.96}$ over a factor 8 in intensity (linear theory: I^1). The transfer is therefore linear in the drive to within the measurement, at zero field and at 10 MG alike. The measured pump-band energy change across the interaction region has median magnitude 2% and maximum 7% over the new cells, i.e. at or below the 5% threshold declared in advance for pump depletion, and it does not grow with intensity. The driven density perturbation is linear in intensity too ($I:\delta n/n = 0.25:0.0009, 0.50:0.0017, 1.00:0.0029, 2.00:0.0055$). Saturation — pump depletion, ion trapping, any mechanism nonlinear in the drive — is excluded as the reason the absolute model over-predicts, because a saturated transfer would not scale linearly. Both pre-registered nonlinear candidates, C2 and V2, are refuted directly by this arm rather than merely left unattributed.

Temperature. Panel (c) moves T_e over a factor of three at fixed T_e/T_i , geometry and intensity, so ZT_e/T_i — and with it the ion Landau damping and Q — is unchanged ($Q = 52, 53$). This separates T_e from Q , which the published data confounds, and the answer is unambiguous: the measured gain barely moves while the model's absolute prediction moves by a factor of three, and stays one to two orders of magnitude above the measurement at every temperature. $B = 0$ MG: $g(0.4 \text{ keV}) = 0.28 \pm 0.10$ against 28.6 absolute and 0.47 registered, $g(0.8 \text{ keV}) = 0.23 \pm 0.09$ against 13.3 absolute and 0.23 registered, $g(1.2 \text{ keV}) = 0.18 \pm 0.06$ against 8.4 absolute and 0.15 registered; $B = 10$ MG: $g(0.4 \text{ keV}) = 0.24 \pm 0.09$ against 21.7 absolute and 0.33 registered, $g(0.8 \text{ keV}) = 0.21 \pm 0.10$ against 10.8 absolute and 0.18 registered, $g(1.2 \text{ keV}) = 0.17 \pm 0.06$ against 7.1 absolute and 0.12 registered. Every cell breaks, at every temperature, exactly as a Q -controlled boundary requires and as a T_e -controlled one would not.

Orientation. Two orientations the published set does not cover were added, and they test the boundary rather than merely extend the grid: at $\theta_{kB} \simeq 25^\circ$ magnetisation barely moves Q , so the model should still fail at every field, while at $\theta_{kB} \simeq 70^\circ$ it moves Q down but — on the prediction made from the published cells — not far enough to cross Q^* even at 50 MG. Both expectations are borne out. $\theta_{kB} \simeq 70^\circ$: $g(0 \text{ MG}) = 0.23 \pm 0.09$ vs model 13.27, $Q = 52.9$, $g(10 \text{ MG}) = 0.21 \pm 0.09$ vs model 5.45, $Q = 25.3$, $g(50 \text{ MG}) = 0.12 \pm 0.06$ vs model 1.72, $Q = 18.0$; $\theta_{kB} \simeq 25^\circ$: $g(0 \text{ MG}) = 0.23 \pm 0.09$ vs model 13.27, $Q = 52.9$, $g(10 \text{ MG}) = 0.21 \pm 0.06$ vs model 12.62, $Q = 50.7$, $g(50 \text{ MG}) = 0.13 \pm 0.05$ vs model 9.99, $Q = 47.5$. Orientation therefore does not act on the verdict except through Q : $\theta_{kB} = 70^\circ$ at 50 MG breaks while $\theta_{kB} = 88^\circ$ at the same field holds, and the two differ in Q by an order of magnitude.

Background offset. The declared rule was that the non-CBET background scales with interaction length, $g_0(200\lambda_0) = 0.4 g_0(500\lambda_0) = -0.157$. The measurement is $g_0 = -0.013$, a factor 12 smaller than the length-scaled expectation of -0.157 . The declared linear scaling of the background with interaction length is therefore wrong: the non-CBET loss falls much faster than the interaction length, which is consistent with it being dominated by instabilities that need many growth lengths. The measured value is used, and the discrepancy is reported here because the rule was declared in advance.

Numerical checks. At the reference point ($\theta_{kB} = 45^\circ$, $B = 10$ MG, parallel polarisation), changing the resolution over 12, 15, 20 cells per λ_0 moves the gain by at most 6% of its value ($g = 0.221$, $g = 0.209$, $g = 0.217$); extending the domain to $300\lambda_0$ gives $g = 0.455$ against 0.209 at $200\lambda_0$; and the run-to-run scatter over 3 independent PIC noise realisations at 3 cells is $\sigma_{\text{seed}} = 0.008$. These bracket the reduction and set the noise floor used in every new-run uncertainty.

An independent check that does not use the gain. The low-frequency density perturbation at the beat wavevector is measurable directly. In the unmagnetised reduced-scale run it rises from $\delta n/n = 0.0001$ before the pump arrives to 0.0017 in quasi-steady state, while the linear response function predicts 0.0245 at the same detuning — a factor 15 larger. The response function itself, not the way it is turned into a gain, is what is over-predicted.

5.10 The confirmatory arm at intermediate damping

The confirmatory cells place Q between the two clusters the published data samples, in the same plasma, geometry, intensity and electron temperature as the reference, with only the ion temperature changed. Their absolute predictions were recorded before the simulations were run. $ZT_e/T_i = 9$ ($Q = 16.1$), $B = 0$ MG: predicted 2.50, measured 0.21 ± 0.10 , $D = 10.07$ (breaks); $ZT_e/T_i = 9$ ($Q = 15.3$), $B = 10$ MG: predicted 2.22, measured 0.20 ± 0.09 , $D = 8.93$ (breaks); $ZT_e/T_i = 6$ ($Q = 7.2$), $B = 0$ MG: predicted 0.96, measured 0.19 ± 0.10 , $D = 3.38$ (breaks); $ZT_e/T_i = 6$ ($Q = 7.0$), $B = 10$ MG: predicted 0.87, measured 0.19 ± 0.10 , $D = 3.01$ (breaks). Every one of them still breaks under the verdict rule — the reduced-scale family is judged on a small gain scale — but the over-prediction factor falls smoothly and by a large factor as Q falls ($Q = 52.9$: $S = 37$, $Q = 16.1$: $S = 9$, $Q = 7.2$: $S = 5$), which is the transition the classification split stands in for, measured on cells that took no part in locating it. These four cells are also the ones that fix the exponent of Eq. (4) in the region where the published data has none.

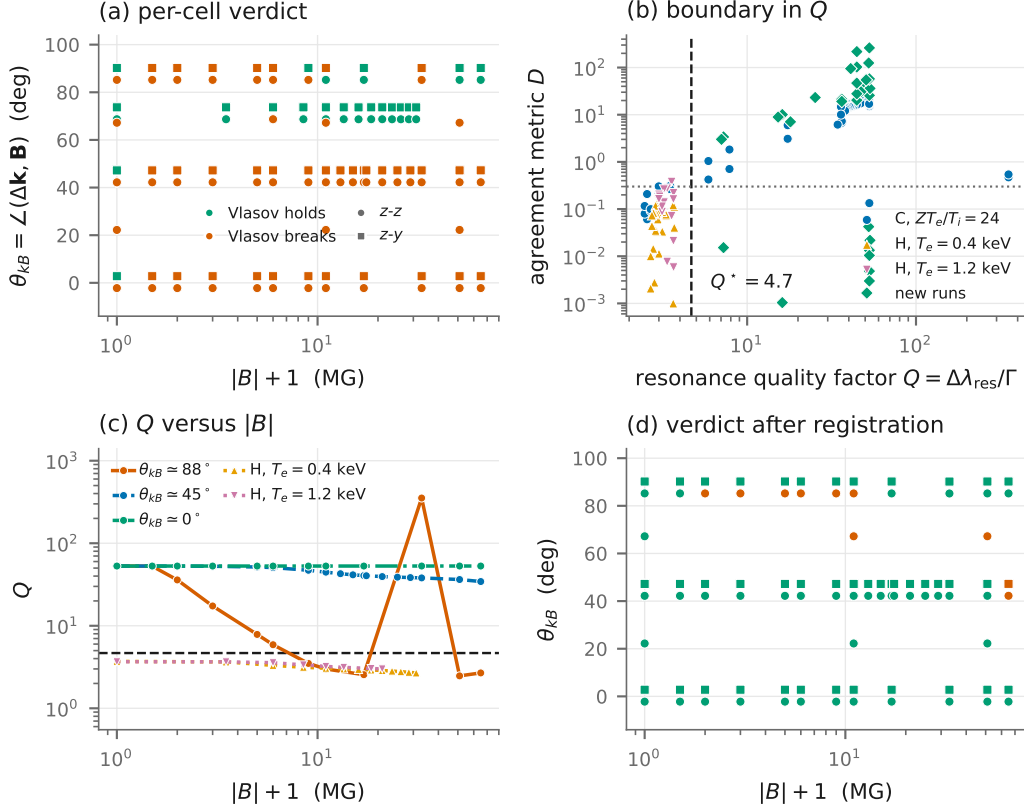


Figure 1: The validity boundary. (a) Per-cell verdict for the absolute model in the plane of field strength and mediating-wavevector orientation; circles are parallel, squares orthogonal polarisation. (b) The same verdicts against the quality factor Q of the mediating resonance; the dashed line is the single split that classifies 91% of cells, the dotted line is the verdict threshold $D = 0.3$. (c) Q versus field strength: magnetisation lowers Q only when $\Delta\mathbf{k}$ is close to perpendicular to $\mathbf{B}$; the spike at 32 MG is the emergence of a narrow magnetised mode. (d) Verdict for the registered model in the carbon family.

6 Failure analysis

Where the classification is fuzzy. The verdict rule normalises by a fixed gain scale, which is what makes it usable for the orthogonal-polarisation cells but also means that a cell in which both model and measurement are small can be recorded as holding even when they differ by a large factor. This affects the $\theta_{kB} \approx 88^\circ$ carbon cells above 16 MG in the registered map, where a measured 0.35 and a modelled 0.03 give $D = 0.21$. We flag those cells rather than claim them: they are consistent with agreement but do not demonstrate it. The three $B = 0$ orthogonal-polarisation cells, where the model predicts exactly zero and the measurement returns the residual background 0.20, are in the same category.

The new cells are judged on a harder scale. The verdict rule normalises by the unmagnetised gain of the same family, and the reduced-scale family has $g_{\text{ref}} = 0.23$ against 1.51 for the published full-scale carbon runs. The same absolute discrepancy therefore produces a ~ 6 times larger D at the reduced scale. This is the metric as declared and we have not softened it, but it means the new-run verdicts are systematically more demanding than the provided-data verdicts, and a new-run cell recorded as breaking is not directly comparable to a provided-data cell recorded as breaking.

Mechanisms that did not work. Two of the seven pre-registered indicators, C3 and C4, never exceeded their thresholds anywhere in the swept region and so could never be attributed. V2's ion

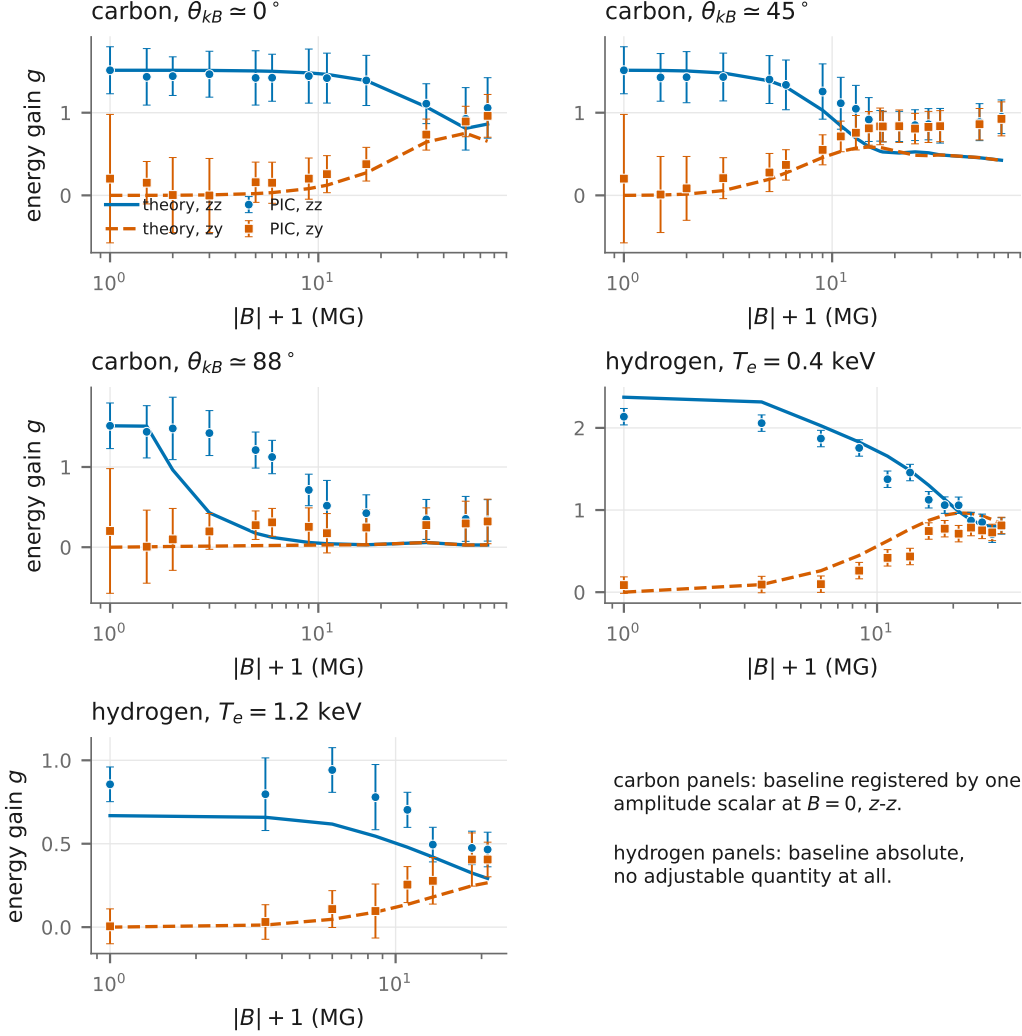


Figure 2: Measured (points, with t_1-t_2 and f_y uncertainty) and modelled (lines) energy gain against field strength, for parallel (z-z) and orthogonal (z-y) initial polarisations. Carbon panels use the registered model; hydrogen panels use the absolute model, with nothing adjusted.

term — the one the hypothesis actually names — stays at 4×10^{-4} of the ion temperature even at $2 \times 10^{15} \text{ W cm}^{-2}$, because it is suppressed by $Z^2 m_e/m_i$; the indicator crosses only through its electron term. That is a useful negative result: the ion-ponderomotive correction is not what limits linear CBET modelling at ICF intensities, and it should be dropped from the candidate list in any follow-up. V1 produced the cleanest null: no association at all, and its indicator is diagnostically awkward besides, since it diverges at $B = 0$ where the term it bounds vanishes identically (28 cells).

Axes that turned out to be inert. Field strength and polarisation, the two axes the magnetised problem is usually organised around, carry almost none of the agreement signal on their own. This is not to say magnetisation is unimportant — it changes the gain by a factor of three — but that the model tracks those changes about as well or as badly as it tracks the unmagnetised case, once Q is fixed.

Thresholds tried and rejected. We did not tune the verdict threshold; the declared value and its two-point robustness band are reported together and the classification is insensitive across them. What we did revise, and disclose, is the baseline: the plane-wave absolute form was found to be far outside

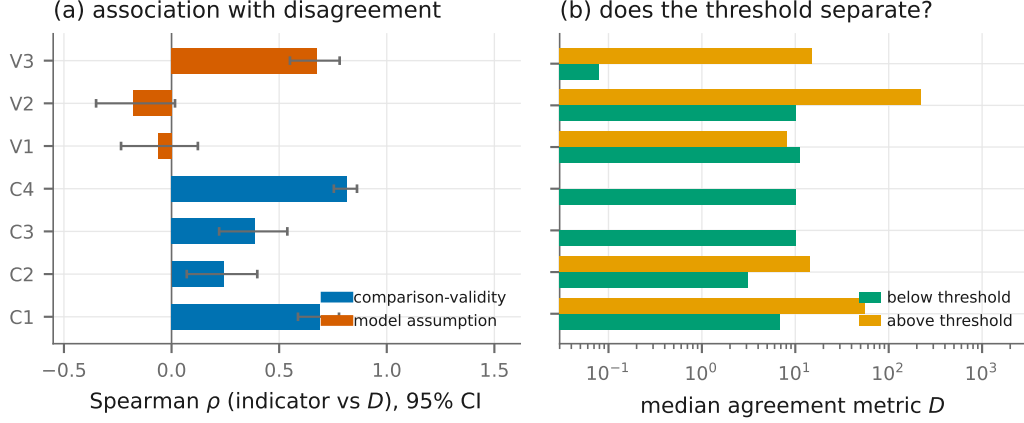


Figure 3: Pre-registered failure modes. (a) Rank correlation of each indicator with the disagreement metric, with bootstrap 95% intervals; blue are comparison-validity prerequisites, vermilion are physical assumptions of the model. (b) Median disagreement below and above each indicator’s *a priori* threshold; a bar pair is only evidence of a mechanism if both bars exist and differ.

the linear regime at the carbon reference point, and rather than abandon it we report it alongside a one-parameter registered form. The pre-declared V3 cut at $Q > 3$ is reported as insufficiently discriminating rather than quietly moved.

A residual we did not explain. The resonance-position offset of Sec. 5.7 is the largest single discrepancy that survives everything else, and we could not attribute it. Distinguishing trapping-induced detuning from beam-geometry broadening requires a Fourier-resolved diagnostic of the driven ion-acoustic wave in (ω, k) , which the decks used here do not output; we record it as open.

7 Related work and what is new here

Validation of reduced CBET models against kinetic reference data is an established activity, and it is worth being precise about which part of it this paper adds to. Viala *et al.* [9] validate a linear kinetic ion-acoustic CBET model against ignition-scale polar-direct-drive experiments, and earlier work established the model at OMEGA scale; those studies test a model against measurements at a design point. Seaton *et al.* [14] and Yin *et al.* [11] characterise the nonlinear and kinetic corrections to unmagnetised CBET with PIC and with reduced nonlinear models, Nguyen *et al.* [10] identify trapping-induced detuning as a saturation route, Hansen *et al.* [12] identify ion heating and pump depletion, and Turnbull *et al.* [13] identify the Langdon effect on the damping. Each of these isolates one mechanism. Lalaire *et al.* [15] show that optical smoothing broadens the CBET resonance well beyond its plane-wave width, which is directly relevant to the resonance-position and width discrepancies reported in Sec. 5.7. On the magnetised side, the theory of magnetised three-wave coupling and magnetised LPI has been developed in a series of papers [23, 24, 25, 29, 31, 30], and a linearised Vlasov treatment of magnetised CBET specifically now exists [26] with PIC support [27].

What is new here is the framing as a *validity map with a controlling coordinate*. Rather than asking how large a particular correction is at a design point, we ask which region of a five-parameter space a reduced model can be used in at all, fix the verdict rule before looking, and let the data choose the coordinate the region is simplest in. That coordinate turns out to be the damping of the mediating wave, which unifies effects that the mechanism-by-mechanism literature treats separately: ion Landau damping (set by ZT_e/T_i), magnetisation-induced cyclotron phase mixing (set by θ_{kB} and $|B|$), and the saturation route (weak damping $\Rightarrow$ resonant gain outside the linear regime $\Rightarrow$ trapping and depletion). We are not aware of a published boundary-identification rule for CBET models of the form used here, and our verdict rule is deliberately plain — a normalised residual with a pre-declared threshold and an uncertainty test — rather than a novel statistical method; the multiple-comparison machinery (bootstrap rank correlations with Benjamini–Hochberg control [35]) is standard and is

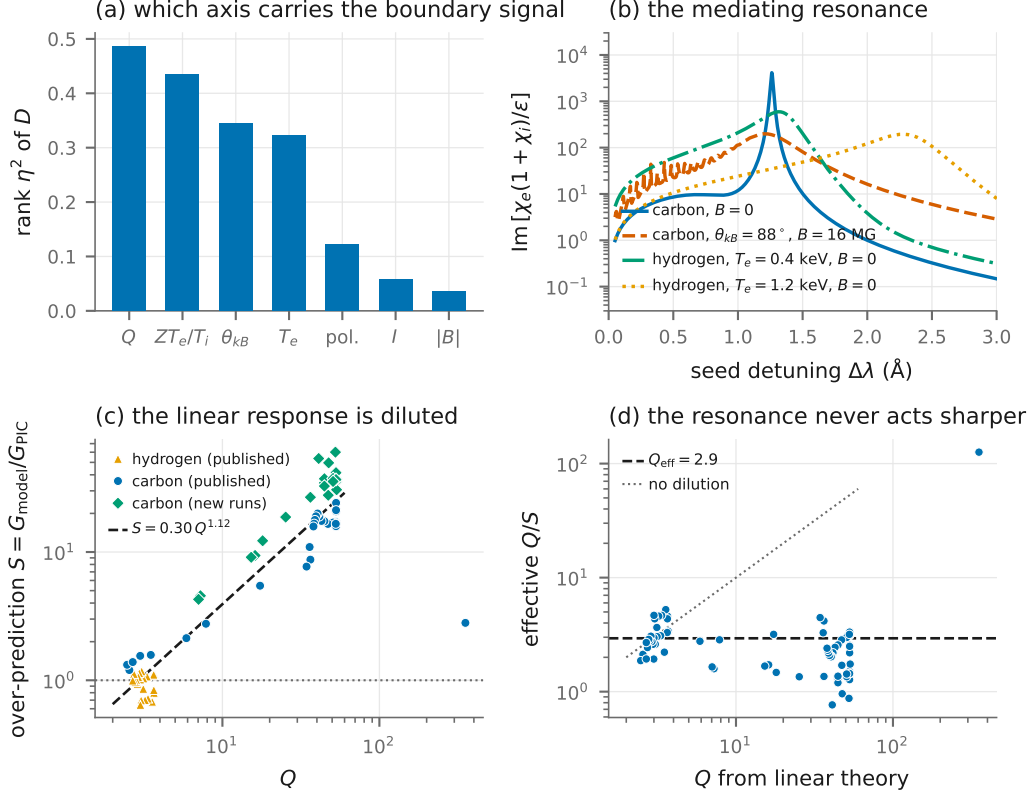


Figure 4: (a) Fraction of the rank variance of the disagreement metric explained by each swept axis. (b) The mediating low-frequency response $\text{Im } X$ against seed detuning for four representative conditions.

used as such. The reduced-scale PIC validation follows the ordinary practice of checking a cheaper configuration against a published reference and against resolution and length variation; no part of it is claimed as new.

8 Computational cost

The model is cheap: one sweep cell costs 0.2 s of a single core, dominated by the resonance search, and the whole 160-cell map plus all mechanism indicators takes under two minutes. The kinetic reference is not: the new reduced-scale simulations cost a median 21 minutes each on 32 cores (3000×150 cells, 7.2×10^6 particles, 27 789 steps, i.e. 10.9 core-hours per run), and the published full-scale runs they are reduced from carry roughly $8 \times$ more work each. The whole new-simulation campaign reported here is 41 runs and 509 core-hours.

The relevant question for extending the map is what five more simulations would buy. Along the axes that turned out inert — field strength, polarisation — almost nothing; those cells are already dense in the published data. Along Q they would buy a great deal, because the published set samples Q at only two clusters, near 3 and near 50, and the split at $Q^* = 4.7$ is therefore located by a gap rather than by a transition. Five simulations at intermediate ZT_e/T_i — for instance a CH or beryllium plasma at $ZT_e/T_i \simeq 8\text{--}12$ — would place Q between the clusters and convert a classification boundary into a measured transition. That is the single most valuable extension, and it is cheap: those runs need no new machinery.

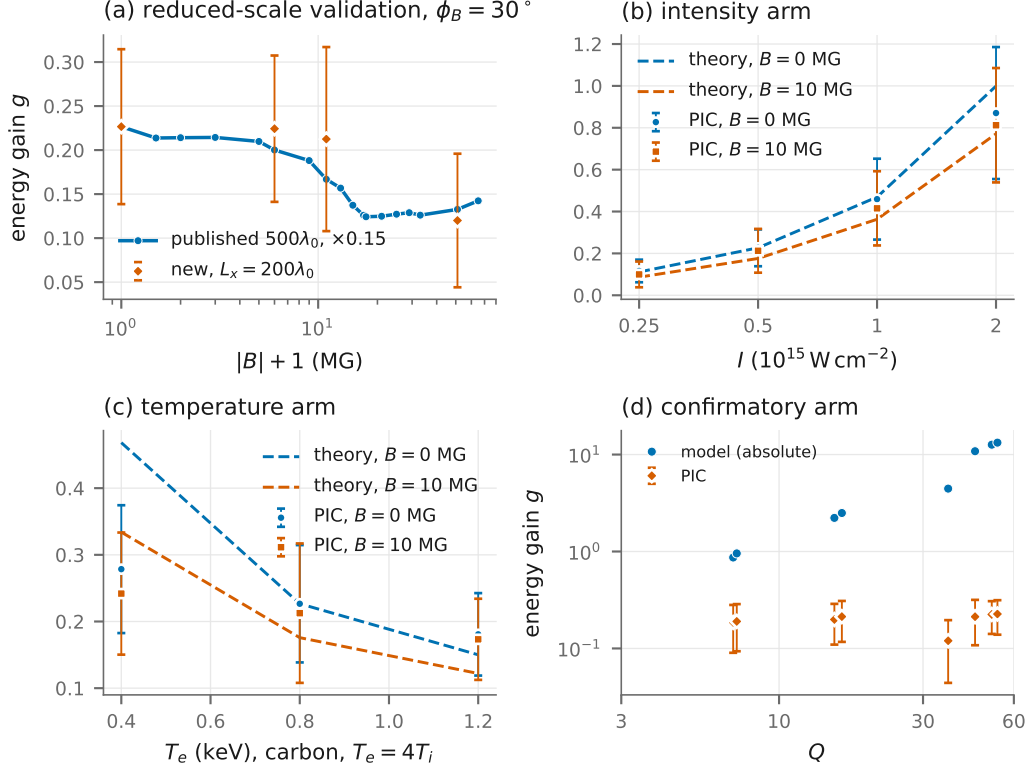


Figure 5: New reduced-scale simulations. (a) Reduced-scale reproduction of the published field scan at $\theta_{kB} = 45^\circ$, after a single amplitude rescaling. (b) Intensity arm. (c) Temperature arm. (d) The confirmatory arm: measured and *absolutely* predicted gain against the mediating-resonance quality factor, on cells whose ion temperature was raised to place Q between the two clusters the published data samples.

9 Implications for practice

The result translates into a pre-shot test that costs one susceptibility evaluation. Compute $X(\Delta\omega, \Delta\mathbf{k})$ from Eq. (1) on a detuning scan, take $Q = \Delta\lambda_{\text{res}}/\Gamma$, and:

- **$Q \lesssim 5$: use the reduced model.** Strongly damped mediators occur at low ZT_e/T_i — hydrogen, deuterium, DT, or any plasma with hot ions — and in near-perpendicular magnetised geometries. Here the linear magnetised model reproduced measured gains to a median 7–16% of the unmagnetised gain scale with nothing fitted, including the magnetised polarisation-conversion channel. Ray-based CBET packages using linear kinetic response functions are on solid ground in this regime, magnetised or not.
- **A correction, not just a veto.** Between the two regimes the over-prediction is not arbitrary: Eq. (4) gives it as $S \simeq Q/Q_{\text{eff}}$ with $Q_{\text{eff}} = 3.0$ at the published interaction length, over a decade and a half in Q and across two ion species, both polarisations and 0–64 MG. A practitioner who must use the linear model outside its region can divide the predicted amplitude gain by $\max(1, Q/3.0)$ and expect to land within a factor of about two, which is far better than the factor of 37 the raw model is off by at the carbon reference point. This is an empirical correction fitted to 86 cells, not a derivation, and it should be treated as such.
- **$Q \gtrsim 10$: do not trust the raw absolute gain.** Weakly damped mediators occur at high ZT_e/T_i — mid- Z ablator and hohlraum-wall plasmas with cold ions are the obvious ICF case — and there the linear resonant gain is so large that the transfer is saturation-limited in reality while the model reports an unbounded resonance. The model’s absolute output should be treated as an upper bound only. Its *relative* predictions — how the gain changes

with field strength, orientation and polarisation — remain usable after a single calibration at one measured operating point, which is how it should be embedded in a design code.

- **Trust the amplitude, not the location.** Even inside the trustworthy region the model puts the resonance at a detuning 1.42–1.82 too large (Sec. 5.7). A design that uses a reduced model to choose the wavelength separation between beam cones is therefore using the least reliable output of that model: the peak height transfers, the peak position does not. Wavelength-detuning schemes should be set from measurements or from a model that carries the nonlinear frequency shift, not from the linear root alone.
- **Magnetisation is not the risk factor.** Over 0–64 MG the field never made the reduced model worse in the regime where it was already good. Where the field acts on the mediating wave (near-perpendicular $\Delta\mathbf{k}$) it broadens the resonance and improves the model. Design teams adding an imposed field to a configuration whose CBET modelling already validates should not expect the magnetisation itself to invalidate it.
- **Watch the perpendicular corner at high field.** The one place the field introduced a genuinely new failure is the emergence of narrow magnetised resonances at $\theta_{kB} \rightarrow 90^\circ$ and tens of MG, which no single-mediator model represents. This is a small corner of parameter space, but it is exactly the corner that near-perpendicular imposed-field geometries occupy.

10 Limitations

The kinetic reference is 2D-3V and collisionless, so collisional damping — which raises Q^{-1} and would therefore move conditions *into* the trustworthy region — is absent from both sides of the comparison; the boundary reported here is conservative in that respect. The reduced-scale simulations are 0.4 of the published interaction length and 0.75 of its resolution. Resolution is converged — the gain varies by 6% over 12, 15 and 20 cells per λ_0 — but length is not a free reduction: the amplitude gain scales as $L^{1.83}$ over the range tested, so the new runs sit at a systematically stronger dilution ($Q_{\text{eff}} = 1.9$) than the published configuration (3.0), and every new-run verdict is referred to its own family’s unmagnetised gain for that reason. The suppression law is an empirical fit over two interaction lengths only, and its length dependence is constrained by a single pair of runs. The model’s absolute normalisation and its resonance position are both known to be affected by physics outside its scope, and we register only the first of these. The confirmatory arm fills the gap between the two Q clusters the published data samples, but with four cells only, all in one plasma composition. Finally, the attribution rests on a pre-registered menu of seven mechanisms; a failure mode outside that menu would be recorded here as “V3” if it happened to co-vary with the resonance width.

11 Conclusion

The question “when can a linearised Vlasov model be trusted for magnetised cross-beam energy transfer?” has a compact answer that is not about the magnetic field. It is about whether the wave doing the mediating is strongly damped. Across 160 kinetic reference points spanning field strength, orientation, temperature, intensity and polarisation, a single derived coordinate — the quality factor of the mediating resonance — classifies 91% of the verdicts, and the assumption that fails at the boundary is the one that puts the mediator there. Magnetisation matters to this picture mainly as a knob that can move the damping, and it moves it in the helpful direction.

Data, code and reproducibility

All code is in `proposal/code/`: the pre-registered plan and its dated amendments, the model, the analysis-chain port, the simulation driver and manifest, the sweep evaluation and the figure and table generators. All logs are in `proposal/logs/`: per-run configuration and wall clock, per-run gains and diagnostics, the per-cell sweep table in JSON and CSV, and the mechanism and compactness summaries. Every number quoted in this paper is generated from those logs by `make_tables.py`. Inspection notes on the reused packages are in `proposal/codebase_notes.md` and the record of what was tried and abandoned is in `proposal/attempts_log.md`.

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A Complete per-cell table

Every sweep cell with a logged kinetic measurement and a model evaluation appears below; none is omitted. The machine-readable version is `proposal/logs/sweep_cells.csv`.

#	parameters	model g	PIC $g \pm \sigma$	D	verdict	mech.
1	0 MG, -13° , $C_{3.5}$, 0.8, 0.50, zy	0.00	0.20 ± 0.78	0.13	holds	n/a
2	0 MG, -13° , $C_{3.5}$, 0.8, 0.50, zy	25.73	0.01 ± 0.46	17.00	breaks	unattributed
3	1 MG, -13° , $C_{3.5}$, 0.8, 0.50, zy	15.32	0.10 ± 0.39	10.06	breaks	V3
4	2 MG, -13° , $C_{3.5}$, 0.8, 0.50, zy	4.88	0.20 ± 0.22	3.09	breaks	V3
5	4 MG, -13° , $C_{3.5}$, 0.8, 0.50, zy	1.34	0.27 ± 0.18	0.71	breaks	V3
6	5 MG, -13° , $C_{3.5}$, 0.8, 0.50, zy	0.95	0.31 ± 0.17	0.42	breaks	V3
7	8 MG, -13° , $C_{3.5}$, 0.8, 0.50, zy	0.66	0.25 ± 0.24	0.27	holds	n/a
8	10 MG, -13° , $C_{3.5}$, 0.8, 0.50, zy	0.63	0.17 ± 0.25	0.30	breaks	V3
9	16 MG, -13° , $C_{3.5}$, 0.8, 0.50, zy	0.56	0.25 ± 0.22	0.21	holds	n/a
10	32 MG, -13° , $C_{3.5}$, 0.8, 0.50, zy	1.10	0.28 ± 0.21	0.54	breaks	V3
11	50 MG, -13° , $C_{3.5}$, 0.8, 0.50, zy	0.47	0.30 ± 0.28	0.12	holds	n/a
12	64 MG, -13° , $C_{3.5}$, 0.8, 0.50, zy	0.47	0.32 ± 0.28	0.10	holds	n/a
13	0 MG, -13° , $C_{3.5}$, 0.8, 0.50, zz	36.42	1.51 ± 0.28	23.06	breaks	C2
14	0 MG, -13° , $C_{3.5}$, 0.8, 0.50, zz	36.35	1.44 ± 0.33	23.07	breaks	C2
15	1 MG, -13° , $C_{3.5}$, 0.8, 0.50, zz	23.77	1.48 ± 0.39	14.73	breaks	C2
16	2 MG, -13° , $C_{3.5}$, 0.8, 0.50, zz	10.48	1.42 ± 0.28	5.99	breaks	C2
17	4 MG, -13° , $C_{3.5}$, 0.8, 0.50, zz	3.98	1.21 ± 0.22	1.83	breaks	C2
18	5 MG, -13° , $C_{3.5}$, 0.8, 0.50, zz	2.70	1.12 ± 0.21	1.04	breaks	C2
19	8 MG, -13° , $C_{3.5}$, 0.8, 0.50, zz	1.18	0.71 ± 0.20	0.31	breaks	V3
20	10 MG, -13° , $C_{3.5}$, 0.8, 0.50, zz	0.83	0.52 ± 0.32	0.21	holds	n/a
21	16 MG, -13° , $C_{3.5}$, 0.8, 0.50, zz	0.52	0.42 ± 0.23	0.06	holds	n/a
22	32 MG, -13° , $C_{3.5}$, 0.8, 0.50, zz	1.06	0.35 ± 0.25	0.47	breaks	V3
23	50 MG, -13° , $C_{3.5}$, 0.8, 0.50, zz	0.47	0.35 ± 0.28	0.08	holds	n/a
24	64 MG, -13° , $C_{3.5}$, 0.8, 0.50, zz	0.47	0.33 ± 0.26	0.09	holds	n/a
25	0 MG, $+30^\circ$, $C_{3.5}$, 0.8, 0.50, zy	0.00	0.20 ± 0.78	0.13	holds	n/a
26	0 MG, $+30^\circ$, $C_{3.5}$, 0.8, 0.50, zy	25.54	0.01 ± 0.46	16.87	breaks	unattributed
27	1 MG, $+30^\circ$, $C_{3.5}$, 0.8, 0.50, zy	26.89	0.08 ± 0.39	17.71	breaks	V3
28	2 MG, $+30^\circ$, $C_{3.5}$, 0.8, 0.50, zy	28.09	0.21 ± 0.25	18.42	breaks	V3
29	4 MG, $+30^\circ$, $C_{3.5}$, 0.8, 0.50, zy	28.79	0.28 ± 0.23	18.84	breaks	V3
30	5 MG, $+30^\circ$, $C_{3.5}$, 0.8, 0.50, zy	28.60	0.37 ± 0.18	18.65	breaks	V3
31	8 MG, $+30^\circ$, $C_{3.5}$, 0.8, 0.50, zy	26.86	0.55 ± 0.18	17.38	breaks	V3
32	10 MG, $+30^\circ$, $C_{3.5}$, 0.8, 0.50, zy	25.48	0.71 ± 0.19	16.36	breaks	V3
33	12 MG, $+30^\circ$, $C_{3.5}$, 0.8, 0.50, zy	24.39	0.76 ± 0.21	15.61	breaks	V3
34	14 MG, $+30^\circ$, $C_{3.5}$, 0.8, 0.50, zy	23.59	0.81 ± 0.20	15.05	breaks	V3
35	16 MG, $+30^\circ$, $C_{3.5}$, 0.8, 0.50, zy	23.04	0.83 ± 0.22	14.67	breaks	C2
36	16 MG, $+30^\circ$, $C_{3.5}$, 0.8, 0.50, zy	22.92	0.83 ± 0.19	14.59	breaks	C2
37	20 MG, $+30^\circ$, $C_{3.5}$, 0.8, 0.50, zy	22.14	0.84 ± 0.20	14.07	breaks	C2
38	24 MG, $+30^\circ$, $C_{3.5}$, 0.8, 0.50, zy	21.33	0.81 ± 0.18	13.56	breaks	V3
39	28 MG, $+30^\circ$, $C_{3.5}$, 0.8, 0.50, zy	20.38	0.83 ± 0.19	12.92	breaks	C2
40	32 MG, $+30^\circ$, $C_{3.5}$, 0.8, 0.50, zy	19.16	0.84 ± 0.19	12.11	breaks	C2
41	50 MG, $+30^\circ$, $C_{3.5}$, 0.8, 0.50, zy	11.90	0.86 ± 0.19	7.29	breaks	C2
42	64 MG, $+30^\circ$, $C_{3.5}$, 0.8, 0.50, zy	10.26	0.93 ± 0.21	6.17	breaks	C2
43	0 MG, $+30^\circ$, $C_{3.5}$, 0.8, 0.50, zz	36.42	1.51 ± 0.28	23.06	breaks	C2
44	0 MG, $+30^\circ$, $C_{3.5}$, 0.8, 0.50, zz	36.40	1.43 ± 0.29	23.11	breaks	C2
45	1 MG, $+30^\circ$, $C_{3.5}$, 0.8, 0.50, zz	36.36	1.43 ± 0.31	23.08	breaks	C2
46	2 MG, $+30^\circ$, $C_{3.5}$, 0.8, 0.50, zz	36.16	1.43 ± 0.29	22.95	breaks	C2
47	4 MG, $+30^\circ$, $C_{3.5}$, 0.8, 0.50, zz	35.43	1.40 ± 0.29	22.49	breaks	C2

#	parameters	model g	PIC $g \pm \sigma$	D	verdict	mech.
48	5 MG, +30°, C _{3.5} , 0.8, 0.50, zz	34.77	1.34 ± 0.30	22.09	breaks	C2
49	8 MG, +30°, C _{3.5} , 0.8, 0.50, zz	31.92	1.26 ± 0.33	20.26	breaks	C2
50	10 MG, +30°, C _{3.5} , 0.8, 0.50, zz	29.98	1.11 ± 0.31	19.07	breaks	C2
51	12 MG, +30°, C _{3.5} , 0.8, 0.50, zz	28.43	1.05 ± 0.28	18.09	breaks	C2
52	14 MG, +30°, C _{3.5} , 0.8, 0.50, zz	27.24	0.92 ± 0.27	17.39	breaks	C2
53	16 MG, +30°, C _{3.5} , 0.8, 0.50, zz	26.36	0.84 ± 0.18	16.86	breaks	C2
54	16 MG, +30°, C _{3.5} , 0.8, 0.50, zz	26.16	0.83 ± 0.18	16.73	breaks	C2
55	20 MG, +30°, C _{3.5} , 0.8, 0.50, zz	24.89	0.83 ± 0.18	15.90	breaks	C2
56	24 MG, +30°, C _{3.5} , 0.8, 0.50, zz	23.61	0.85 ± 0.19	15.04	breaks	C2
57	28 MG, +30°, C _{3.5} , 0.8, 0.50, zz	22.22	0.86 ± 0.19	14.11	breaks	C2
58	32 MG, +30°, C _{3.5} , 0.8, 0.50, zz	20.56	0.84 ± 0.21	13.03	breaks	C2
59	50 MG, +30°, C _{3.5} , 0.8, 0.50, zz	11.08	0.89 ± 0.22	6.73	breaks	C2
60	64 MG, +30°, C _{3.5} , 0.8, 0.50, zz	10.33	0.95 ± 0.20	6.20	breaks	C2
61	0 MG, +75°, C _{3.5} , 0.8, 0.50, zy	0.00	0.20 ± 0.78	0.13	holds	n/a
62	0 MG, +75°, C _{3.5} , 0.8, 0.50, zy	23.14	0.15 ± 0.26	15.19	breaks	unattributed
63	1 MG, +75°, C _{3.5} , 0.8, 0.50, zy	24.53	0.00 ± 0.45	16.20	breaks	V3
64	2 MG, +75°, C _{3.5} , 0.8, 0.50, zy	25.92	-0.00 ± 0.45	17.12	breaks	V3
65	4 MG, +75°, C _{3.5} , 0.8, 0.50, zy	27.30	0.16 ± 0.24	17.93	breaks	V3
66	5 MG, +75°, C _{3.5} , 0.8, 0.50, zy	27.74	0.15 ± 0.25	18.23	breaks	V3
67	8 MG, +75°, C _{3.5} , 0.8, 0.50, zy	28.68	0.20 ± 0.25	18.81	breaks	V3
68	10 MG, +75°, C _{3.5} , 0.8, 0.50, zy	29.12	0.26 ± 0.22	19.07	breaks	V3
69	16 MG, +75°, C _{3.5} , 0.8, 0.50, zy	30.03	0.38 ± 0.20	19.59	breaks	V3
70	32 MG, +75°, C _{3.5} , 0.8, 0.50, zy	31.26	0.74 ± 0.19	20.17	breaks	V3
71	50 MG, +75°, C _{3.5} , 0.8, 0.50, zy	31.84	0.89 ± 0.18	20.44	breaks	C2
72	64 MG, +75°, C _{3.5} , 0.8, 0.50, zy	31.93	0.96 ± 0.26	20.46	breaks	C2
73	0 MG, +75°, C _{3.5} , 0.8, 0.50, zz	36.42	1.51 ± 0.28	23.06	breaks	C2
74	0 MG, +75°, C _{3.5} , 0.8, 0.50, zz	36.42	1.43 ± 0.34	23.12	breaks	C2
75	1 MG, +75°, C _{3.5} , 0.8, 0.50, zz	36.42	1.44 ± 0.23	23.11	breaks	C2
76	2 MG, +75°, C _{3.5} , 0.8, 0.50, zz	36.42	1.47 ± 0.28	23.09	breaks	C2
77	4 MG, +75°, C _{3.5} , 0.8, 0.50, zz	36.42	1.42 ± 0.33	23.12	breaks	C2
78	5 MG, +75°, C _{3.5} , 0.8, 0.50, zz	36.41	1.42 ± 0.28	23.12	breaks	C2
79	8 MG, +75°, C _{3.5} , 0.8, 0.50, zz	36.41	1.44 ± 0.33	23.10	breaks	C2
80	10 MG, +75°, C _{3.5} , 0.8, 0.50, zz	36.40	1.42 ± 0.30	23.11	breaks	C2
81	16 MG, +75°, C _{3.5} , 0.8, 0.50, zz	36.36	1.39 ± 0.30	23.11	breaks	C2
82	32 MG, +75°, C _{3.5} , 0.8, 0.50, zz	36.18	1.11 ± 0.24	23.17	breaks	C2
83	50 MG, +75°, C _{3.5} , 0.8, 0.50, zz	35.81	0.93 ± 0.38	23.05	breaks	C2
84	64 MG, +75°, C _{3.5} , 0.8, 0.50, zz	35.36	1.06 ± 0.37	22.66	breaks	C2
85	0 MG, +30°, C _{3.5} , 0.4, 0.50, zy	0.00	0.01 ± 0.01	0.04	holds	n/a
86	0 MG, +30°, C _{3.5} , 0.4, 0.50, zz	28.57	0.28 ± 0.10	124.86	breaks	C1
87	10 MG, +30°, C _{3.5} , 0.4, 0.50, zz	21.74	0.24 ± 0.09	94.89	breaks	C1
88	0 MG, +30°, C _{3.5} , 0.8, 0.25, zz	5.94	0.12 ± 0.05	25.70	breaks	unattributed
89	10 MG, +30°, C _{3.5} , 0.8, 0.25, zz	4.72	0.10 ± 0.06	20.39	breaks	V3
90	0 MG, +30°, C _{3.5} , 0.8, 0.50, zy	0.00	-0.00 ± 0.02	0.02	holds	n/a
91	0 MG, +30°, C _{3.5} , 0.8, 0.50, zz	0.96	0.19 ± 0.10	3.38	breaks	C1
92	10 MG, +30°, C _{3.5} , 0.8, 0.50, zz	0.87	0.19 ± 0.10	3.01	breaks	C1
93	0 MG, +30°, C _{3.5} , 0.8, 0.50, zy	0.00	0.00 ± 0.02	0.00	holds	n/a
94	0 MG, +30°, C _{3.5} , 0.8, 0.50, zz	2.50	0.21 ± 0.10	10.07	breaks	C1
95	10 MG, +30°, C _{3.5} , 0.8, 0.50, zz	2.22	0.20 ± 0.09	8.93	breaks	C1
96	0 MG, +5°, C _{3.5} , 0.8, 0.50, zz	13.27	0.23 ± 0.09	57.56	breaks	C1
97	10 MG, +5°, C _{3.5} , 0.8, 0.50, zz	5.45	0.21 ± 0.09	23.15	breaks	C1
98	50 MG, +5°, C _{3.5} , 0.8, 0.50, zz	1.72	0.12 ± 0.06	7.07	breaks	V3
99	0 MG, +30°, C _{3.5} , 0.8, 0.50, zy	0.00	0.00 ± 0.02	0.00	holds	n/a
100	0 MG, +30°, C _{3.5} , 0.8, 0.50, zy	0.00	-0.00 ± 0.03	0.01	holds	n/a
101	0 MG, +30°, C _{3.5} , 0.8, 0.50, zy	0.00	0.00 ± 0.02	0.01	holds	n/a
102	10 MG, +30°, C _{3.5} , 0.8, 0.50, zy	6.31	0.01 ± 0.02	27.82	breaks	V3
103	50 MG, +30°, C _{3.5} , 0.8, 0.50, zy	4.95	0.12 ± 0.07	21.29	breaks	V3
104	0 MG, +30°, C _{3.5} , 0.8, 0.50, zz	13.27	0.23 ± 0.09	57.56	breaks	C1
105	5 MG, +30°, C _{3.5} , 0.8, 0.50, zz	12.64	0.22 ± 0.08	54.78	breaks	C1
106	10 MG, +30°, C _{3.5} , 0.8, 0.50, zz	10.82	0.21 ± 0.10	46.83	breaks	C1
107	50 MG, +30°, C _{3.5} , 0.8, 0.50, zz	4.45	0.12 ± 0.08	19.12	breaks	V3
108	0 MG, +50°, C _{3.5} , 0.8, 0.50, zz	13.27	0.23 ± 0.09	57.56	breaks	C1
109	10 MG, +50°, C _{3.5} , 0.8, 0.50, zz	12.62	0.21 ± 0.06	54.74	breaks	V3
110	50 MG, +50°, C _{3.5} , 0.8, 0.50, zz	9.99	0.13 ± 0.05	43.54	breaks	V3
111	0 MG, +30°, C _{3.5} , 0.8, 1.00, zz	28.62	0.46 ± 0.19	124.28	breaks	unattributed
112	10 MG, +30°, C _{3.5} , 0.8, 1.00, zz	23.71	0.42 ± 0.18	102.77	breaks	unattributed
113	0 MG, +30°, C _{3.5} , 0.8, 2.00, zy	0.00	0.00 ± 0.02	0.00	holds	n/a

#	parameters	model g	PIC $g \pm \sigma$	D	verdict	mech.
114	0 MG, +30°, C _{3.5} , 0.8, 2.00, zz	60.02	0.87 ± 0.32	261.02	breaks	unattributed
115	10 MG, +30°, C _{3.5} , 0.8, 2.00, zz	50.11	0.81 ± 0.27	217.53	breaks	unattributed
116	0 MG, +30°, C _{3.5} , 1.2, 0.50, zy	0.00	−0.00 ± 0.04	0.02	holds	n/a
117	0 MG, +30°, C _{3.5} , 1.2, 0.50, zz	8.37	0.18 ± 0.06	36.14	breaks	unattributed
118	10 MG, +30°, C _{3.5} , 1.2, 0.50, zz	7.11	0.17 ± 0.06	30.62	breaks	V3
119	0 MG, +15°, H _{3.0} , 0.4, 0.50, zy	0.00	0.09 ± 0.10	0.04	holds	n/a
120	2 MG, +15°, H _{3.0} , 0.4, 0.50, zy	0.09	0.09 ± 0.10	0.00	holds	n/a
121	5 MG, +15°, H _{3.0} , 0.4, 0.50, zy	0.26	0.10 ± 0.10	0.08	holds	n/a
122	8 MG, +15°, H _{3.0} , 0.4, 0.50, zy	0.45	0.26 ± 0.10	0.09	holds	n/a
123	10 MG, +15°, H _{3.0} , 0.4, 0.50, zy	0.62	0.42 ± 0.10	0.10	holds	n/a
124	12 MG, +15°, H _{3.0} , 0.4, 0.50, zy	0.77	0.43 ± 0.10	0.16	holds	n/a
125	15 MG, +15°, H _{3.0} , 0.4, 0.50, zy	0.87	0.74 ± 0.10	0.06	holds	n/a
126	18 MG, +15°, H _{3.0} , 0.4, 0.50, zy	0.94	0.77 ± 0.10	0.08	holds	n/a
127	20 MG, +15°, H _{3.0} , 0.4, 0.50, zy	0.97	0.71 ± 0.10	0.12	holds	n/a
128	22 MG, +15°, H _{3.0} , 0.4, 0.50, zy	0.96	0.79 ± 0.10	0.08	holds	n/a
129	25 MG, +15°, H _{3.0} , 0.4, 0.50, zy	0.93	0.75 ± 0.10	0.08	holds	n/a
130	28 MG, +15°, H _{3.0} , 0.4, 0.50, zy	0.88	0.73 ± 0.10	0.07	holds	n/a
131	30 MG, +15°, H _{3.0} , 0.4, 0.50, zy	0.83	0.81 ± 0.10	0.01	holds	n/a
132	0 MG, +15°, H _{3.0} , 0.4, 0.50, zz	2.37	2.14 ± 0.10	0.11	holds	n/a
133	2 MG, +15°, H _{3.0} , 0.4, 0.50, zz	2.32	2.06 ± 0.10	0.12	holds	n/a
134	5 MG, +15°, H _{3.0} , 0.4, 0.50, zz	2.03	1.87 ± 0.10	0.07	holds	n/a
135	8 MG, +15°, H _{3.0} , 0.4, 0.50, zz	1.83	1.76 ± 0.10	0.03	holds	n/a
136	10 MG, +15°, H _{3.0} , 0.4, 0.50, zz	1.66	1.37 ± 0.10	0.13	holds	n/a
137	12 MG, +15°, H _{3.0} , 0.4, 0.50, zz	1.48	1.46 ± 0.10	0.01	holds	n/a
138	15 MG, +15°, H _{3.0} , 0.4, 0.50, zz	1.30	1.12 ± 0.10	0.08	holds	n/a
139	18 MG, +15°, H _{3.0} , 0.4, 0.50, zz	1.13	1.06 ± 0.10	0.03	holds	n/a
140	20 MG, +15°, H _{3.0} , 0.4, 0.50, zz	0.98	1.06 ± 0.10	0.03	holds	n/a
141	22 MG, +15°, H _{3.0} , 0.4, 0.50, zz	0.88	0.87 ± 0.10	0.00	holds	n/a
142	25 MG, +15°, H _{3.0} , 0.4, 0.50, zz	0.82	0.85 ± 0.10	0.01	holds	n/a
143	28 MG, +15°, H _{3.0} , 0.4, 0.50, zz	0.80	0.71 ± 0.10	0.04	holds	n/a
144	30 MG, +15°, H _{3.0} , 0.4, 0.50, zz	0.80	0.81 ± 0.10	0.00	holds	n/a
145	0 MG, +15°, H _{3.0} , 1.2, 0.50, zy	0.00	0.00 ± 0.10	0.01	holds	n/a
146	2 MG, +15°, H _{3.0} , 1.2, 0.50, zy	0.01	0.03 ± 0.10	0.02	holds	n/a
147	5 MG, +15°, H _{3.0} , 1.2, 0.50, zy	0.05	0.11 ± 0.11	0.07	holds	n/a
148	8 MG, +15°, H _{3.0} , 1.2, 0.50, zy	0.09	0.10 ± 0.16	0.01	holds	n/a
149	10 MG, +15°, H _{3.0} , 1.2, 0.50, zy	0.14	0.26 ± 0.11	0.14	holds	n/a
150	12 MG, +15°, H _{3.0} , 1.2, 0.50, zy	0.18	0.28 ± 0.14	0.11	holds	n/a
151	18 MG, +15°, H _{3.0} , 1.2, 0.50, zy	0.25	0.41 ± 0.16	0.18	holds	n/a
152	20 MG, +15°, H _{3.0} , 1.2, 0.50, zy	0.27	0.41 ± 0.10	0.16	holds	n/a
153	0 MG, +15°, H _{3.0} , 1.2, 0.50, zz	0.67	0.86 ± 0.10	0.22	holds	n/a
154	2 MG, +15°, H _{3.0} , 1.2, 0.50, zz	0.66	0.80 ± 0.22	0.16	holds	n/a
155	5 MG, +15°, H _{3.0} , 1.2, 0.50, zz	0.62	0.94 ± 0.13	0.38	breaks	C2
156	8 MG, +15°, H _{3.0} , 1.2, 0.50, zz	0.55	0.78 ± 0.20	0.27	holds	n/a
157	10 MG, +15°, H _{3.0} , 1.2, 0.50, zz	0.48	0.70 ± 0.11	0.26	holds	n/a
158	12 MG, +15°, H _{3.0} , 1.2, 0.50, zz	0.42	0.49 ± 0.10	0.09	holds	n/a
159	18 MG, +15°, H _{3.0} , 1.2, 0.50, zz	0.32	0.48 ± 0.10	0.18	holds	n/a
160	20 MG, +15°, H _{3.0} , 1.2, 0.50, zz	0.29	0.47 ± 0.10	0.20	holds	n/a

Table 1: Representative sweep cells (70 of 160; the complete table is in Appendix A and in the logged `sweep_cells.csv`). Parameters are ($|B|$, ϕ_B , $n_e[10^{20} \text{ cm}^{-3}]$, $T_e[\text{keV}]$, $I[10^{15} \text{ W cm}^{-2}]$, pol.); species is C or H. “Model g ” is the absolute form. Mechanism labels are C1–C4 (comparison-
validity) and V1–V3 (physical assumptions) as defined in Sec. 4.

#	parameters	model g	PIC $g \pm \sigma$	D	verdict	mechanism
1	0 MG, +75°, C _{3.5} , 0.8, 0.50, zz	36.42	1.51 ± 0.28	23.06	breaks	C2
2	0 MG, +75°, C _{3.5} , 0.8, 0.50, zy	0.00	0.20 ± 0.78	0.13	holds	n/a
3	2 MG, +75°, C _{3.5} , 0.8, 0.50, zz	36.42	1.47 ± 0.28	23.09	breaks	C2
4	2 MG, +75°, C _{3.5} , 0.8, 0.50, zy	25.92	−0.00 ± 0.45	17.12	breaks	V3
5	10 MG, +75°, C _{3.5} , 0.8, 0.50, zz	36.40	1.42 ± 0.30	23.11	breaks	C2
6	10 MG, +75°, C _{3.5} , 0.8, 0.50, zy	29.12	0.26 ± 0.22	19.07	breaks	V3
7	64 MG, +75°, C _{3.5} , 0.8, 0.50, zz	35.36	1.06 ± 0.37	22.66	breaks	C2
8	64 MG, +75°, C _{3.5} , 0.8, 0.50, zy	31.93	0.96 ± 0.26	20.46	breaks	C2
9	0 MG, +30°, C _{3.5} , 0.8, 0.50, zz	36.42	1.51 ± 0.28	23.06	breaks	C2
10	0 MG, +30°, C _{3.5} , 0.8, 0.50, zy	0.00	0.20 ± 0.78	0.13	holds	n/a
11	2 MG, +30°, C _{3.5} , 0.8, 0.50, zz	36.16	1.43 ± 0.29	22.95	breaks	C2
12	2 MG, +30°, C _{3.5} , 0.8, 0.50, zy	28.09	0.21 ± 0.25	18.42	breaks	V3
13	10 MG, +30°, C _{3.5} , 0.8, 0.50, zz	29.98	1.11 ± 0.31	19.07	breaks	C2
14	10 MG, +30°, C _{3.5} , 0.8, 0.50, zy	25.48	0.71 ± 0.19	16.36	breaks	V3
15	64 MG, +30°, C _{3.5} , 0.8, 0.50, zz	10.33	0.95 ± 0.20	6.20	breaks	C2
16	64 MG, +30°, C _{3.5} , 0.8, 0.50, zy	10.26	0.93 ± 0.21	6.17	breaks	C2
17	0 MG, −13°, C _{3.5} , 0.8, 0.50, zz	36.42	1.51 ± 0.28	23.06	breaks	C2
18	0 MG, −13°, C _{3.5} , 0.8, 0.50, zy	0.00	0.20 ± 0.78	0.13	holds	n/a
19	2 MG, −13°, C _{3.5} , 0.8, 0.50, zz	10.48	1.42 ± 0.28	5.99	breaks	C2
20	2 MG, −13°, C _{3.5} , 0.8, 0.50, zy	4.88	0.20 ± 0.22	3.09	breaks	V3
21	10 MG, −13°, C _{3.5} , 0.8, 0.50, zz	0.83	0.52 ± 0.32	0.21	holds	n/a
22	10 MG, −13°, C _{3.5} , 0.8, 0.50, zy	0.63	0.17 ± 0.25	0.30	breaks	V3
23	64 MG, −13°, C _{3.5} , 0.8, 0.50, zz	0.47	0.33 ± 0.26	0.09	holds	n/a
24	64 MG, −13°, C _{3.5} , 0.8, 0.50, zy	0.47	0.32 ± 0.28	0.10	holds	n/a
25	0 MG, +15°, H _{3.0} , 0.4, 0.50, zz	2.37	2.14 ± 0.10	0.11	holds	n/a
26	0 MG, +15°, H _{3.0} , 0.4, 0.50, zy	0.00	0.09 ± 0.10	0.04	holds	n/a
27	10 MG, +15°, H _{3.0} , 0.4, 0.50, zz	1.66	1.37 ± 0.10	0.13	holds	n/a
28	10 MG, +15°, H _{3.0} , 0.4, 0.50, zy	0.62	0.42 ± 0.10	0.10	holds	n/a
29	20 MG, +15°, H _{3.0} , 0.4, 0.50, zz	0.98	1.06 ± 0.10	0.03	holds	n/a
30	20 MG, +15°, H _{3.0} , 0.4, 0.50, zy	0.97	0.71 ± 0.10	0.12	holds	n/a
31	0 MG, +15°, H _{3.0} , 1.2, 0.50, zz	0.67	0.86 ± 0.10	0.22	holds	n/a
32	0 MG, +15°, H _{3.0} , 1.2, 0.50, zy	0.00	0.00 ± 0.10	0.01	holds	n/a
33	10 MG, +15°, H _{3.0} , 1.2, 0.50, zz	0.48	0.70 ± 0.11	0.26	holds	n/a
34	10 MG, +15°, H _{3.0} , 1.2, 0.50, zy	0.14	0.26 ± 0.11	0.14	holds	n/a
35	20 MG, +15°, H _{3.0} , 1.2, 0.50, zz	0.29	0.47 ± 0.10	0.20	holds	n/a
36	20 MG, +15°, H _{3.0} , 1.2, 0.50, zy	0.27	0.41 ± 0.10	0.16	holds	n/a
37	0 MG, +30°, C _{3.5} , 0.8, 2.00, zy	0.00	0.00 ± 0.02	0.00	holds	n/a
38	0 MG, +30°, C _{3.5} , 0.8, 0.25, zz	5.94	0.12 ± 0.05	25.70	breaks	unattributed
39	0 MG, +30°, C _{3.5} , 0.8, 1.00, zz	28.62	0.46 ± 0.19	124.28	breaks	unattributed
40	0 MG, +30°, C _{3.5} , 0.8, 2.00, zz	60.02	0.87 ± 0.32	261.02	breaks	unattributed
41	10 MG, +30°, C _{3.5} , 0.8, 0.25, zz	4.72	0.10 ± 0.06	20.39	breaks	V3
42	10 MG, +30°, C _{3.5} , 0.8, 1.00, zz	23.71	0.42 ± 0.18	102.77	breaks	unattributed
43	10 MG, +30°, C _{3.5} , 0.8, 2.00, zz	50.11	0.81 ± 0.27	217.53	breaks	unattributed
44	0 MG, +50°, C _{3.5} , 0.8, 0.50, zz	13.27	0.23 ± 0.09	57.56	breaks	C1
45	10 MG, +50°, C _{3.5} , 0.8, 0.50, zz	12.62	0.21 ± 0.06	54.74	breaks	V3
46	50 MG, +50°, C _{3.5} , 0.8, 0.50, zz	9.99	0.13 ± 0.05	43.54	breaks	V3
47	0 MG, +5°, C _{3.5} , 0.8, 0.50, zz	13.27	0.23 ± 0.09	57.56	breaks	C1
48	10 MG, +5°, C _{3.5} , 0.8, 0.50, zz	5.45	0.21 ± 0.09	23.15	breaks	C1
49	50 MG, +5°, C _{3.5} , 0.8, 0.50, zz	1.72	0.12 ± 0.06	7.07	breaks	V3
50	0 MG, +30°, C _{3.5} , 0.8, 0.50, zy	0.00	−0.00 ± 0.02	0.02	holds	n/a
51	0 MG, +30°, C _{3.5} , 0.8, 0.50, zy	0.00	0.00 ± 0.02	0.00	holds	n/a
52	0 MG, +30°, C _{3.5} , 0.8, 0.50, zz	0.96	0.19 ± 0.10	3.38	breaks	C1
53	0 MG, +30°, C _{3.5} , 0.8, 0.50, zz	2.50	0.21 ± 0.10	10.07	breaks	C1
54	10 MG, +30°, C _{3.5} , 0.8, 0.50, zz	0.87	0.19 ± 0.10	3.01	breaks	C1
55	10 MG, +30°, C _{3.5} , 0.8, 0.50, zz	2.22	0.20 ± 0.09	8.93	breaks	C1
56	0 MG, +30°, C _{3.5} , 0.4, 0.50, zy	0.00	0.01 ± 0.01	0.04	holds	n/a
57	0 MG, +30°, C _{3.5} , 1.2, 0.50, zy	0.00	−0.00 ± 0.04	0.02	holds	n/a
58	0 MG, +30°, C _{3.5} , 0.4, 0.50, zz	28.57	0.28 ± 0.10	124.86	breaks	C1
59	0 MG, +30°, C _{3.5} , 1.2, 0.50, zz	8.37	0.18 ± 0.06	36.14	breaks	unattributed
60	10 MG, +30°, C _{3.5} , 0.4, 0.50, zz	21.74	0.24 ± 0.09	94.89	breaks	C1
61	10 MG, +30°, C _{3.5} , 1.2, 0.50, zz	7.11	0.17 ± 0.06	30.62	breaks	V3
62	0 MG, +30°, C _{3.5} , 0.8, 0.50, zy	0.00	0.00 ± 0.02	0.00	holds	n/a
63	0 MG, +30°, C _{3.5} , 0.8, 0.50, zy	0.00	−0.00 ± 0.03	0.01	holds	n/a
64	0 MG, +30°, C _{3.5} , 0.8, 0.50, zy	0.00	0.00 ± 0.02	0.01	holds	n/a
65	0 MG, +30°, C _{3.5} , 0.8, 0.50, zz	13.27	0.23 ± 0.09	57.56	breaks	C1
66	10 MG, +30°, C _{3.5} , 0.8, 0.50, zy	6.31	0.01 ± 0.02	27.82	breaks	V3
67	10 MG, +30°, C _{3.5} , 0.8, 0.50, zz	10.82	0.21 ± 0.10	46.83	breaks	C1
68	50 MG, +30°, C _{3.5} , 0.8, 0.50, zy	4.95	0.12 ± 0.07	21.29	breaks	V3
69	50 MG, +30°, C _{3.5} , 0.8, 0.50, zz	4.45	0.12 ± 0.08	19.12	breaks	V3